VK FINANCE

Detailed Implementation Document

Architecture, security, data model, APIs and operations

Version 1.1 | 28 October 2026

A source-based guide to the Salesforce finance platform and React Experience Cloud portal.

Includes 12 architecture, use-case, workflow and sequence diagrams, generated role/action tables, REST catalogs and the complete custom-field dictionary.

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1. Purpose, scope and evidence

VK Finance is a multi-family personal finance platform implemented on Salesforce. Families use a React portal on Experience Cloud to manage accounts, income, expenses, transfers, budgets, bills, loans, recurring entries and reports. Internal Salesforce users operate the VK Finance Console to onboard and support families. Portal members are custom records, rather than individual Salesforce users.

This guide describes the checked-in implementation as reviewed on 28 October 2026. It is intended for developers, solution architects, administrators, testers and support owners. Source code and metadata determine the implementation tables. Existing README and security documents provide historical deployment and validation context. No org deployment, live security test or application test was run to prepare this guide.

Document control	Value
Document	VK Finance - Detailed Implementation Document
Edition / date	1.1 / 28 October 2026; expanded diagrams
Scope	Repository implementation, metadata, UI, API, security, operations and handover
Evidence boundary	Local source review; live org state and current scheduled jobs were not independently verified
Outputs	VKFinance_Implementation_Document.docx and VKFinance_Implementation_Document.pdf; Markdown, generator and 12 SVG/PNG diagrams included
Existing design	docs/VKFinance_Solution_Blueprint.pdf is the separate solution blueprint

Source inventory	Count / setting
Apex classes / test-annotated classes	116 / 34 (includes Salesforce template code and test helpers)
Apex REST resource classes	15
Object, event, big-object and custom-metadata definitions	25
Checked-in custom field definitions	248
Role permission records	51
Salesforce source API version	68.0 in sfdx-project.json

Source: README.md; sfdx-project.json; force-app/main/default; docs/Security_Checklist.md.

2. Business capabilities and boundaries

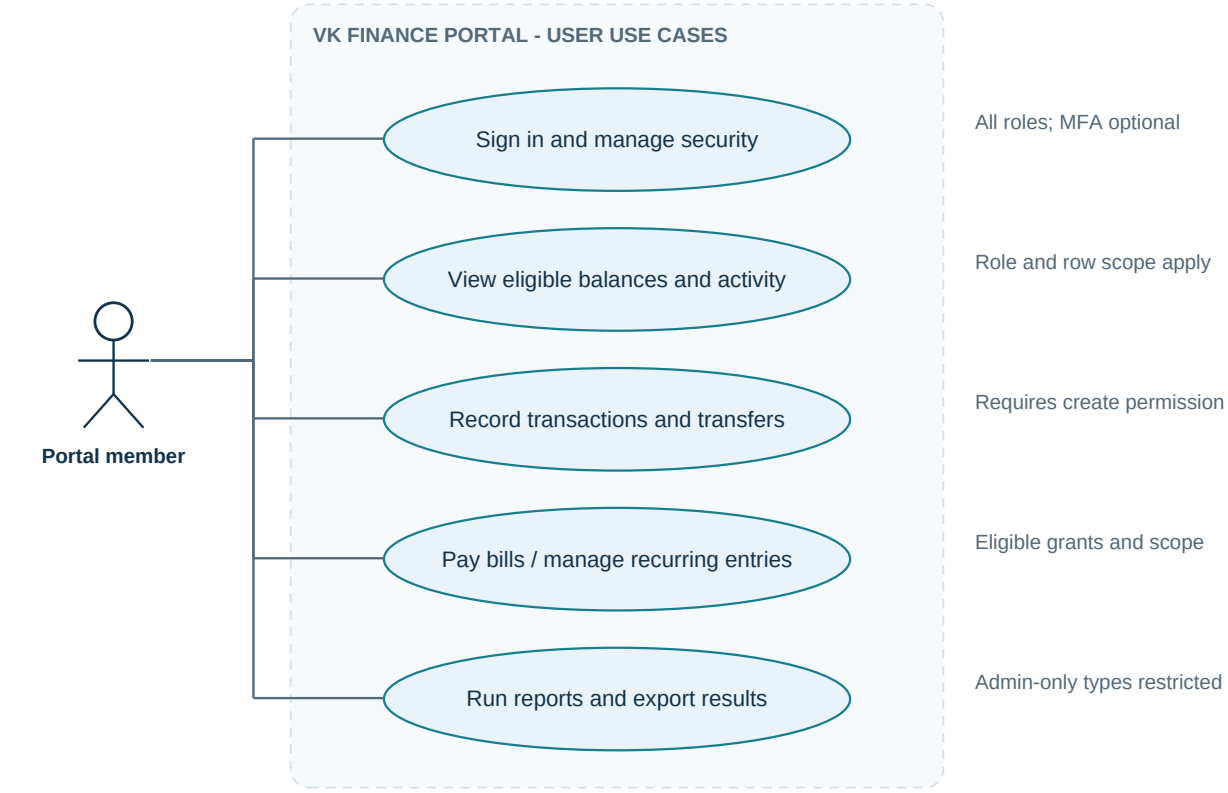
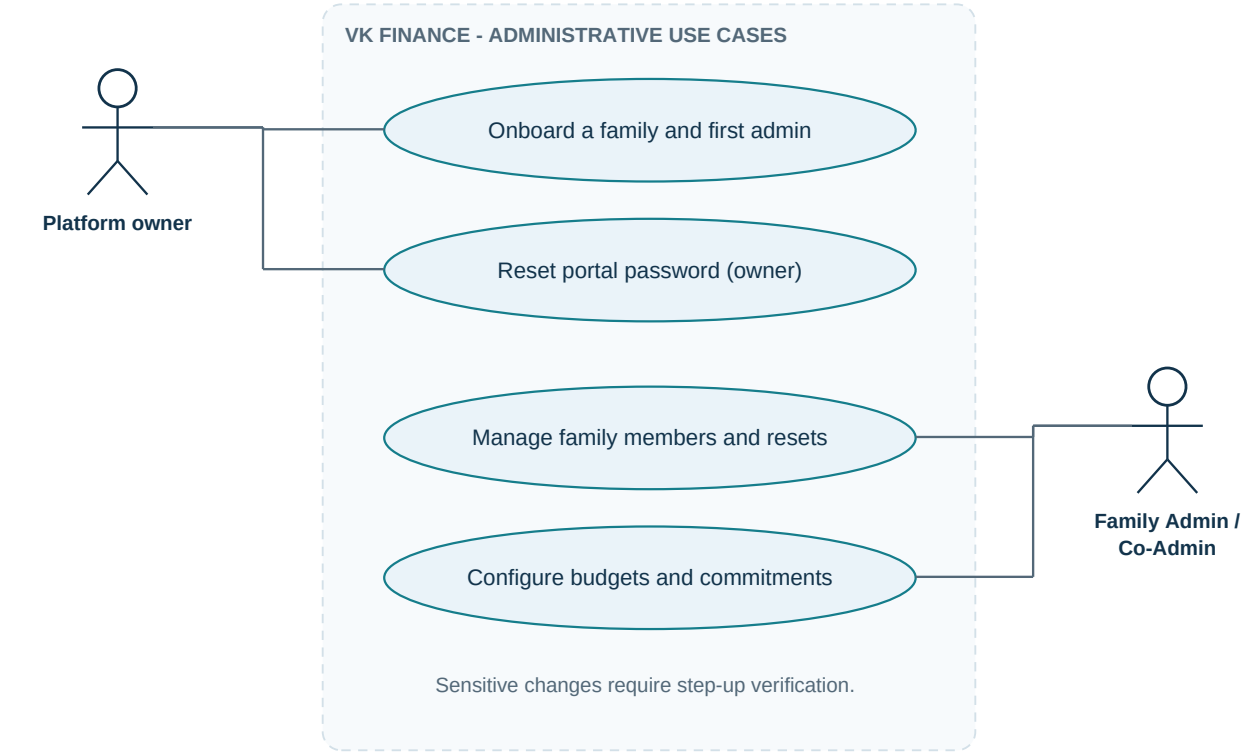


Figure 1. Portal use-case diagram

The portal actor represents authorized family roles. Each use case remains subject to the action grants and row scope described in section 7.

Source: routes.tsx; endpoints.ts; Role_Permission metadata; LedgerScope.cls.



Platform owner uses an internal Salesforce permission. Co-Admin actions retain Family Admin protections.

Figure 2. Administration use-case diagram

The platform owner operates through an internal Salesforce permission. Family administrators operate within one family and retain step-up and Co-Admin restrictions.

Source: VkfConsoleController.cls; FamilyOnboardingService.cls; MemberService.cls.

Capability	Implementation	Result for a family
Identity and membership	AuthService, MemberService, MfaService, FamilyOnboardingService	Invite members, set passwords, manage roles, use authenticator codes and recovery codes.
Core ledger	AccountService, CategoryService, TransactionService, TransferService, LedgerBalances	Record movements and maintain explainable balances across assets and liabilities.
Planning	BudgetService, DashboardService	Plan monthly spending and compare actuals with category budgets.
Commitments	BillService, LoanService, CardStatementService, RecurringService	Track upcoming obligations and record payments through the same ledger.
Automation	DailyDuesBatch, DuesProcessor, NotificationService	Post recurring entries, attempt configured auto-pay, update dues and deliver reminders.
Insights	ReportService, InsightService	Read 13 reports, export CSV, print, save filters and personalize the overview.
Platform operations	VkfConsoleController, scheduled security jobs, AuditService	Onboard families, monitor failures, reconcile balances and control retention.

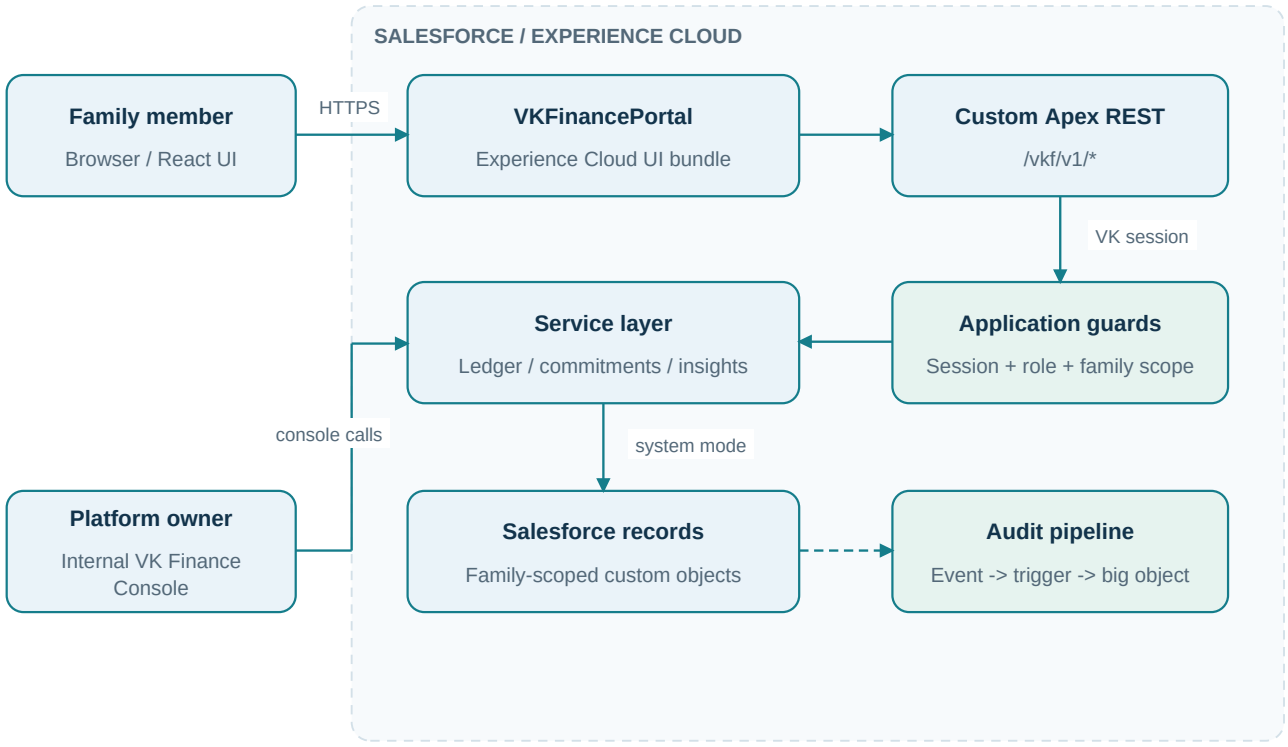
The portal is a bookkeeping application. A payment action records ledger entries; the reviewed implementation does not show bank payment rails or a bank-feed integration. Family base currency and time zone are stored, while reminder money formatting explicitly uses the rupee symbol. Foreign-exchange conversion and fully localized automation must

not be inferred from those configuration fields.

Family onboarding defaults to INR, Asia/Kolkata and 10 members; the service validates a configured member limit between 1 and 50. Categories are seeded from Default_Category__mdt when a family is created. Template Communities controllers also exist in the repository; they are distinct from the custom VK portal identity flow.

Source: FamilyOnboardingService.cls; DuesProcessor.money; package contents; docs/User_Guide.md.

3. Architecture and request lifecycle



Solid arrows: request/data access. Dashed arrow: audit publication (also emitted by guards).

Figure 3. System architecture and execution boundaries

Portal REST requests execute as the site guest user. VK identity and application guards authorize the scoped service access; internal console access has a separate Salesforce permission check.

Source: vkfClient.ts; VkfRest.cls; SessionService.cls; AuditService.cls.

Step / layer	Component	Responsibility
1. Browser	React portal / routes.tsx	Render pages, route authenticated users, show actions from returned permissions.
2. Transport	src/api/vkfClient.ts and endpoints.ts	Use Salesforce Data SDK fetch; send JSON and X-VKF-Session.
3. REST boundary	Vkf*Resource and VkfRest	Dispatch verbs and paths, parse bodies, manage savepoints, normalize responses.
4. Identity	SessionService.require -> UserContext	Validate token, user, family, scope and time limits; load current role from records.
5. Authorization	AccessGuard.require + service checks	Enforce role/feature/action, then business and ownership rules.
6. Data scope	FamilyScopedSelector subclasses + LedgerScope	Constrain family queries and narrow visible accounts and transactions.
7. Persistence	Salesforce custom records + triggers	Use explicit system-mode access after application checks; update ledger balances.

Step / layer	Component	Responsibility
8. Audit / response	Audit_Event__e -> Audit_Log__b; VkfRest.respond	Publish audit events, flush deferred session activity, send uncached DTOs.

The site guest user is the Salesforce execution identity for portal calls. The VK session identifies the actual family member. Services and selectors intentionally use explicit system mode because the guest has no family-data CRUD/FLS access. Consequently, the application-level session, permission and tenant checks are critical security boundaries. Salesforce sharing alone does not implement portal isolation.

Typical write: a resource opens a savepoint, authenticates the VK session, invokes a service, checks a feature action, resolves family-scoped records, applies business validation and saves records. A service returns a DTO rather than serializing unrestricted records. VkfRest sends JSON with Cache-Control: no-store. On failure the REST layer normally rolls back; PortalException.keep() preserves intended security writes such as failed-attempt counters.

Typical read: authenticate, authorize, query the allowed scope, build a DTO and respond. Session activity is deferred until response time so audit big-object reads occur before ordinary DML. New audit reads must preserve this ordering.

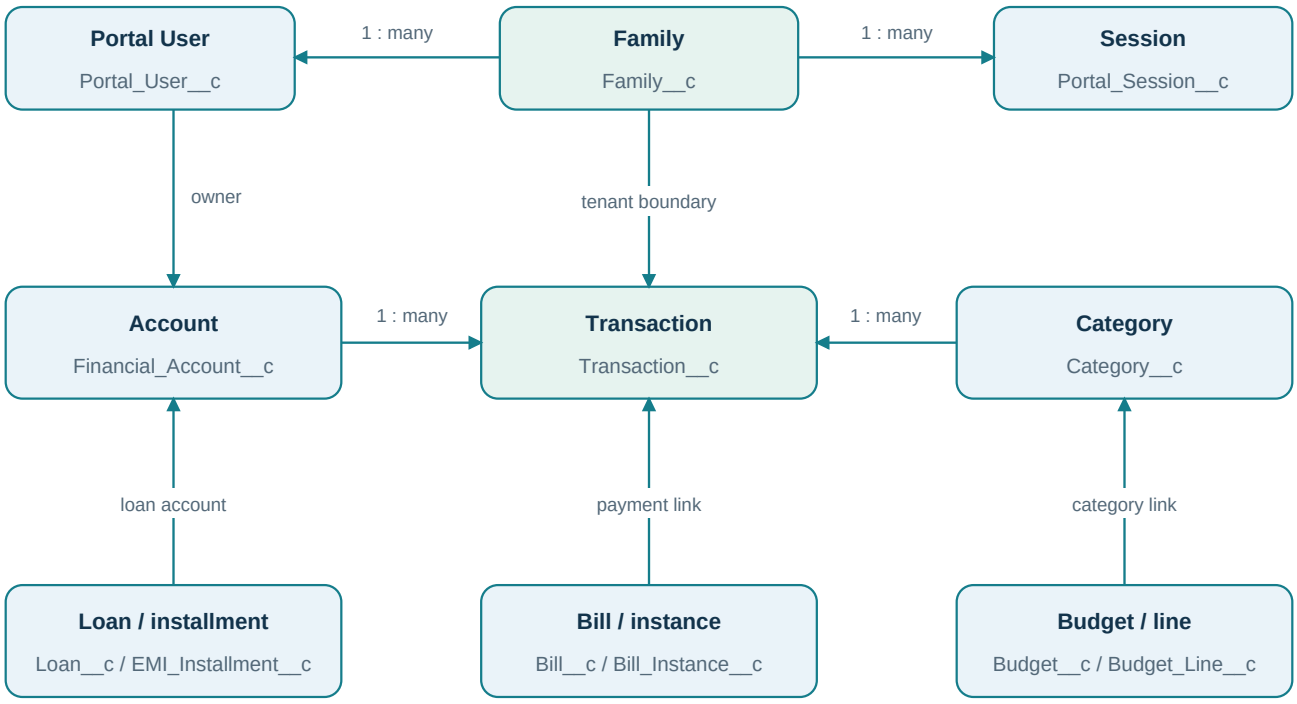
Source: VkfRest.cls; SessionService.cls; AccessGuard.cls; FamilyScopedSelector.cls; FinanceDtos.cls; PortalDtos.cls.

4. Repository and technology map

Location / technology	Purpose / declared version
force-app/main/default/classes	Apex REST, services, selectors, DTOs, jobs and tests.
force-app/main/default/objects	Custom objects, custom metadata, platform event and audit big object.
force-app/main/default/customMetadata	Role grants, default security policy and category templates.
force-app/main/default/uiBundles/VKFinancePortal	React Experience Cloud UI bundle; Node >=22.
React / TypeScript / Vite	Declared React ^19.2.0, TypeScript ~5.9.3 and Vite ^7.2.4.
Salesforce platform SDK / UI bundle	Both declared ^11.57.2; Data SDK transports custom REST calls.
Tailwind / Radix / Zod	Styling, UI primitives and validation; declared ^4.1.17 / ^1.4.3 / ^4.3.5.
Vitest / Playwright	Unit and browser test tooling; declared ^4.1.0 / ^1.49.0.
lwc/vkfFamilyOnboarding	Internal family onboarding interface.
scripts/apex and scripts/security	Scheduling, demo seeding and live black-box security checks.
config/hardening	Scoped Salesforce security settings and guest profile changes.
docs	User guide, security evidence, demo guidance, blueprint and this implementation guide.

Versions above are manifest declarations, not a statement about the currently installed environment. The UI package-lock.json records dependency resolution. Root npm scripts primarily target Salesforce LWC development; portal build and Vitest commands must run inside the UI bundle directory.

5. Data model and relationships



Selected logical relationships. Arrows follow the named relation, not a complete foreign-key or cascade map.
Family predicates also apply to the operational entities; see Appendix A for exact lookup definitions.

Figure 4. Core data relationship diagram

This selected logical view highlights identity, account ownership and ledger links. It omits many tenant links for readability; the field dictionary remains the physical metadata reference.

Source: objects/**/fields/*.field-meta.xml; FamilyScopedSelector.cls.

Entity	Purpose
Audit_Event__e	Immediate-publish event transporting audit facts to the writer.
Audit_Log__b	Indexed audit history held outside normal custom-object storage.
Auth_Token__c	Hashed one-time reset, MFA challenge and recovery-token state.
Bill__c	Reusable bill definition, responsible person and payment/reminder configuration.
Bill_Instance__c	Dated obligation with amounts due, paid and current status.
Budget__c	Monthly family budget header.
Budget_Line__c	Category allocation belonging to a budget.
Card_Statement__c	Card cycle snapshot, statement amount, minimum payment and due date.
Category__c	Family-specific category hierarchy, kinds and system categories.
Default_Category__mdt	Category template copied into each family.

Entity	Purpose
EMI_Installment__c	Persisted loan installment with principal/interest split and due status.
Family__c	Tenant root; family configuration, status, currency, time zone and member cap.
Financial_Account__c	Bank, cash, wallet, card, loan and investment balances and ownership.
Loan__c	Loan terms, borrower, loan account, EMI and outstanding principal.
Monthly_Summary__c	Monthly snapshots supporting older historical trends.
Notification__c	Recipient-specific alert, severity, read state and deduplication key.
Password_History__c	Previous salted hashes used to enforce password reuse policy.
Portal_Session__c	Hashed session token, user/family linkage, scope, activity and revocation.
Portal_User__c	Custom portal identity, family membership, role, password state, MFA and dashboard layout.
Rate_Limit__c	Hashed client-address or throttle bucket and window counters.
Recurring_Rule__c	Recurring transaction configuration, cadence and next-run state.
Role_Permission__mdt	Role x feature x action grants plus descriptive scope.
Saved_Report__c	Report type and saved filters; private or family shared.
Security_Policy__mdt	Default timeouts, password, hashing, throttling and portal URL configuration.
Transaction__c	One ledger movement; linked transfer/payment rows and commitment references.

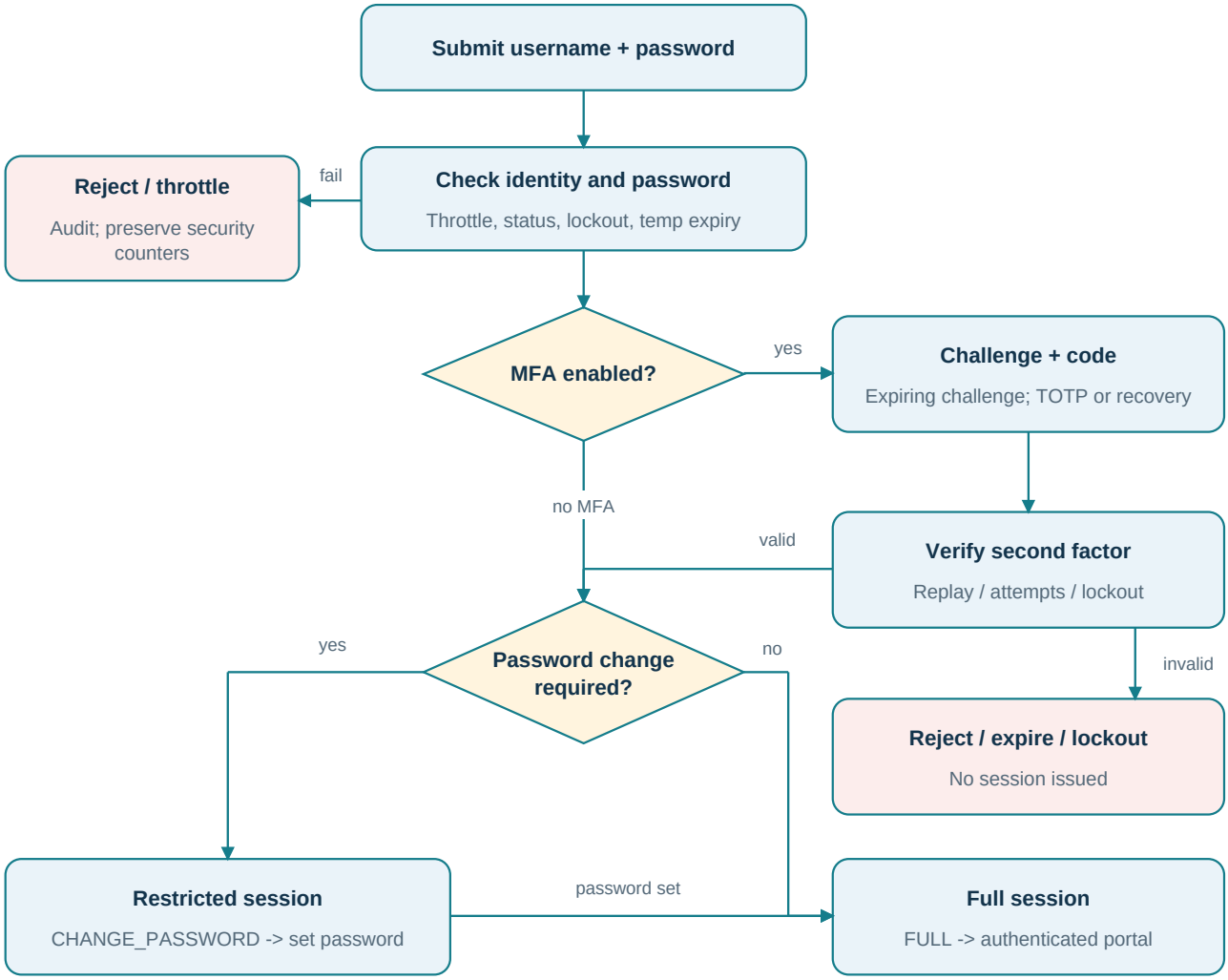
5.1 Relationship interpretation

Family__c is the application tenant. Most operational records carry a direct family reference so selectors can apply a tenant predicate. Some child records reach the family through their parent. A lookup field alone does not prove that both referenced records belong to the same family: services must resolve and validate every client-supplied reference.

Relationship	Implementation meaning
Family -> Portal User -> Session	The session must match the user's current family and active state.
Account -> Transaction -> Category / Member	Transactions explain balance changes and attribution; category and member filters are scoped.
Transaction <-> counter transaction	Transfer_Group_Id__c and Counter_Transaction__c connect related movements.
Budget -> Budget Line -> Category	A month carries planned allocations; actual spending is derived from the ledger.
Bill -> Bill Instance -> Transaction	A recurring definition yields dated obligations; payment rows reference the obligation.
Loan -> EMI Installment -> Transaction	Terms generate installments; principal and interest rows are linked to payment state.
Financial Account -> Card Statement	Statement records are cycle snapshots; payment status is evaluated against ledger credits.
Portal User -> Notification / Saved Report	Recipient and owner scope applies inside the family boundary.

Source: objects/**/fields/*.field-meta.xml; IdentitySelector.cls; CommitmentSelector.cls; TransactionSelector.cls.

6. Authentication, sessions and account security



Simplified decision flow; failed MFA never issues a session. Both login paths can require a password change.

Figure 5. Sign-in, MFA and first-password flowchart

MFA verification precedes session creation when enabled. Both password-only and MFA sign-ins can produce a restricted password-change session; setting the personal password issues full access.

Source: AuthService.login, completeMfa, completeLogin and firstLogin; MfaService.cls.

6.1 Identity flows

Flow	Sequence and persisted effects
Family onboarding	Internal console authorizes the platform owner; creates family, seeded categories and first Family Admin; returns a temporary password once.
First sign-in	Validate temporary credentials; create CHANGE_PASSWORD session; require a new password before full portal access.
Normal sign-in	Normalize username, apply throttling and lockout, verify salted iterated hash; either issue full session or require MFA challenge completion.

Flow	Sequence and persisted effects
MFA sign-in	Validate expiring hashed challenge and TOTP/recovery code; enforce challenge attempts and account lockout; issue full session only after success.
Forgot / reset	Return generic forgot-password response, limit reset emails, validate single-use expiring token and apply password policy.
Password change	Verify current password, enforce length/history, apply new hash and revoke other devices.
Sign-out	Revoke session on the server and clear the browser token.
Member removal	Disable sign-in and clear identity secrets while retaining financial history attribution.

6.2 Security policy defaults

Metadata field	Default	Meaning
Idle_Timeout_Minutes__c	30	Idle session expiration, minutes
Absolute_Timeout_Hours__c	12	Maximum session lifetime, hours
Max_Sessions_Per_User__c	3	Concurrent session cap
Max_Failed_Logins__c	5	Failures before account lockout
Lockout_Minutes__c	30	Initial lockout duration, minutes; repeat lockouts escalate
Throttle_Window_Minutes__c	15	Per-username throttle window, minutes
Throttle_Max_Attempts__c	10	Per-username attempts per window
Min_Password_Length__c	10	Minimum member password length
Min_Admin_Password_Length__c	12	Minimum admin password length
Password_History_Count__c	5	Previous passwords checked
Hash_Iterations__c	3000	Salted HMAC-SHA256 iteration setting
Reset_Token_Minutes__c	30	Password-reset token lifetime, minutes
Temp_Password_Hours__c	24	Temporary password lifetime, hours
Session_Retention_Days__c	7	Closed session retention, days
Portal_Base_Url__c	Environment-specific URL	Email and onboarding destination; change for each environment
MFA_Challenge_Minutes__c	5	MFA login challenge lifetime, minutes
Ip_Max_Failures__c	30	Client-address failure cap per window

Passwords use per-password salt and iterative HMAC-SHA256. Raw session tokens are 256-bit random values and only SHA-256 hashes are persisted. The portal stores the raw token per tab in sessionStorage under vkf.session and sends it in X-VKF-Session. Treat browser script integrity as part of token protection.

SessionService checks active session status, absolute and idle expiration, active user and family, user/session family equality and session scope. A nonblank differing user agent is rejected when a binding hash exists; the code tolerates a missing user-agent header. This is a supplemental check, not a cryptographic device binding.

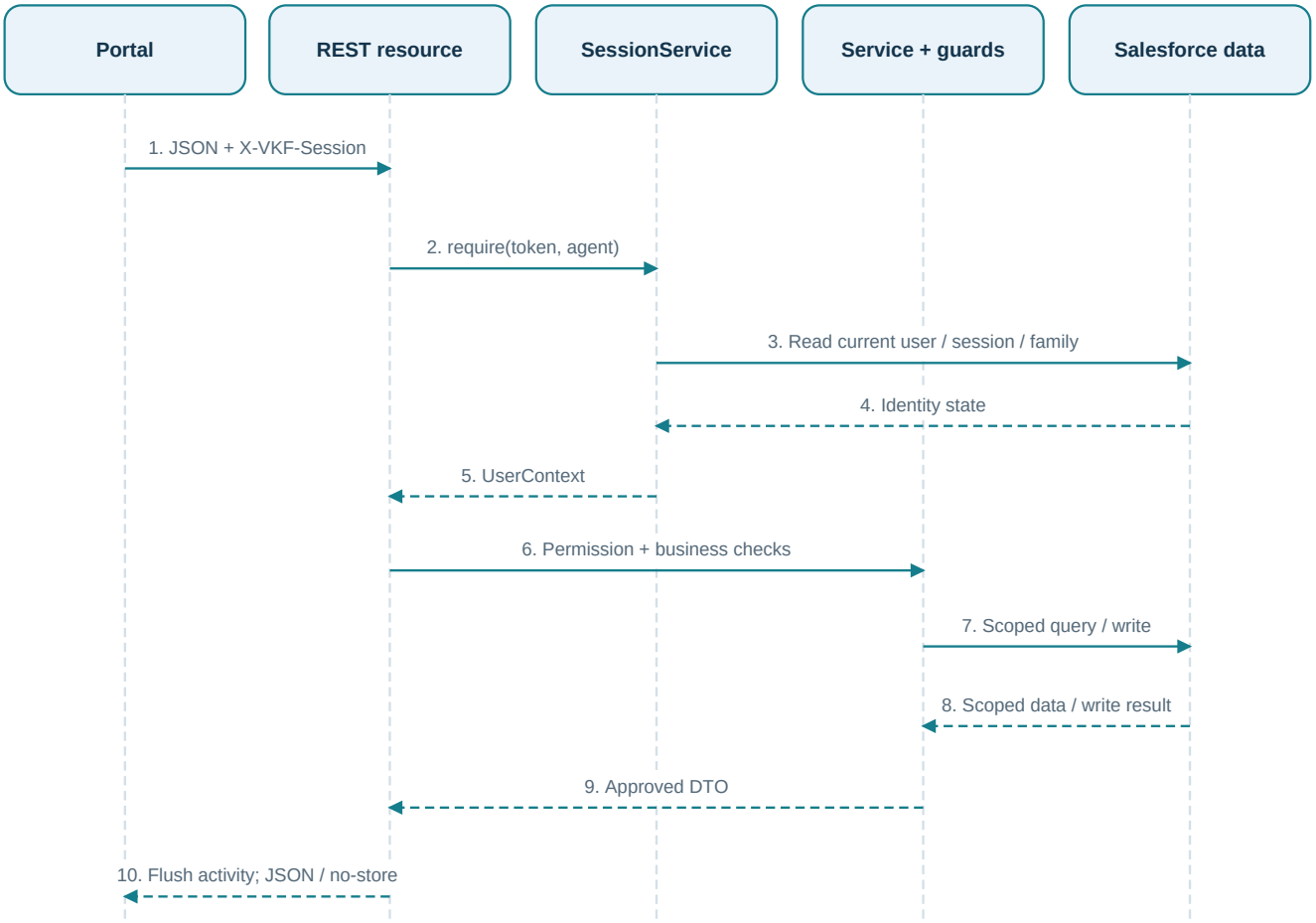
6.3 MFA, recovery and step-up

MfaService implements RFC 6238 authenticator codes, a 30-second period, one window of clock drift and replay protection. The secret is stored in an encrypted field. Setup is confirmed by a valid code. Ten single-use recovery codes are provided; persisted verification state is hashed. Login challenges expire and allow five attempts, with wrong codes also contributing to account lockout.

Sensitive role/status changes, administrative password and MFA resets, MFA disable and recovery-code renewal require step-up verification: current password and, where enabled, an authenticator or recovery code. Failures count toward lockout; lockout revokes sessions. Family-admin MFA reset cannot target the caller or another family. The existing runbook describes platform-owner recovery for a locked-out Family Admin.

Source: AuthService.cls; PasswordService.cls; TokenService.cls; SessionService.cls; MfaService.cls; AccountSecurityService.cls; Security_Policy.Default metadata; src/api/vkfClient.ts.

7. Authorization and family isolation



Write handlers normally use a savepoint. Rejected checks stop before business data access.

Figure 6. Protected API request sequence

The current user and family are resolved before business access. Service permissions and family-scoped selectors determine authorization; safe response construction and deferred activity updates complete the request.

Source: Vkf*Resource.cls; SessionService.require; AccessGuard.cls; VkfRest.respond.

AccessGuard loads Role_Permission__mdt once into a transaction-local map keyed by Role__c + "|" + Feature__c. Actions__c is split on commas, trimmed and uppercased. can() denies a missing grant; require() audits ACCESS_DENIED and throws NOT_AUTHORIZED. permissionsFor() returns sorted FEATURE:ACTION values for UI visibility. Services use uppercase action constants.

Scope__c is not read by AccessGuard. It describes intended scope in metadata; it does not automatically enforce ownership or family visibility. Selectors and service rules implement that scope. Changing a Scope__c value alone does not change record access.

7.1 Exact role/action matrix

Cells below are generated from the checked-in role metadata. A dash means no metadata record or no granted actions. Additional service restrictions continue to apply even when an action is listed. Abbreviations: FA = Family Admin; CA = Co-Admin.

Feature	FA	CA	Member	Child	Viewer
ACCOUNTS	READ, CREATE, EDIT	READ, CREATE, EDIT	READ, CREATE, EDIT	READ	-
BILLS	READ, CREATE, EDIT, PAY	READ, CREATE, EDIT, PAY	READ, PAY	-	READ
BUDGETS	READ, CREATE, EDIT	READ, CREATE, EDIT	READ	-	READ
CATEGORIES	READ, CREATE, EDIT	READ	READ	READ	READ
DASHBOARD	READ	READ	READ	READ	READ
LOANS	READ, CREATE, EDIT, PAY	READ, CREATE, EDIT, PAY	READ	-	READ
MEMBERS	READ, CREATE, EDIT	READ, CREATE, EDIT	READ	READ	READ
MEMBER_SECURITY	EDIT	EDIT	-	-	-
NOTIFICATIONS	READ	READ	READ	READ	READ
RECURRING	READ, CREATE, EDIT	READ, CREATE, EDIT	READ, CREATE, EDIT	-	-
REPORTS	READ, CREATE	READ, CREATE	READ, CREATE	READ, CREATE	READ
TRANSACTIONS	READ, CREATE, EDIT, DELETE	READ, CREATE, EDIT, DELETE	READ, CREATE, EDIT, DELETE	READ, CREATE, EDIT	READ

7.2 Row-level rules

Role / concern	Actual scope rule
Family Admin / Co-Admin	Whole-family ledger visibility. Co-Admin member administration has additional protections around Family Admins.
Member	Own and family-shared accounts; can manage own account settings; may edit own eligible transactions.
Child	Own accounts; own non-void transaction edits only within 24 hours of CreatedDate.
Viewer	Read only; whole-family scope when Viewer_Scope__c = Family; otherwise own-account scope. No posting or account use.
Transaction visibility	Whole-family scope OR attributed to caller OR held on a visible account.
Transfer destination	Any active account in the same family can receive funds; source use is restricted. This supports pocket money to a child's wallet.
Cross-family record IDs	Family-scoped lookup returns NOT_FOUND/404 and can audit CROSS_FAMILY_ATTEMPT.
Notifications	Family and Recipient__c must match the caller. Admins do not receive a family-wide notification feed.

7.3 FA_NOTIFICATIONS worked example

Role_Permission.FA_NOTIFICATIONS.md-meta.xml grants Family Admin the NOTIFICATIONS feature with the READ action and Scope__c = OWN. NotificationService.feed() requires NOTIFICATIONS:READ and reads the caller's latest 30 alerts and unread count. markRead() uses the same READ grant, then filters by family and recipient; an empty ids array means mark the caller's unread alerts read, capped at 2,000 in that branch. No separate EDIT grant is required by this

implementation.

When adding a protected feature, create the intended role grants, call AccessGuard in the service, resolve IDs through scoped selectors, enforce row-level business rules and return a safe DTO. Client route guards help navigation; server checks enforce access.

Source: AccessGuard.cls; LedgerScope.cls; FamilyScopedSelector.cls; MemberService.cls; NotificationService.cls; Role_Permission.FA_NOTIFICATIONS.md-meta.xml.

8. Ledger and planning implementation

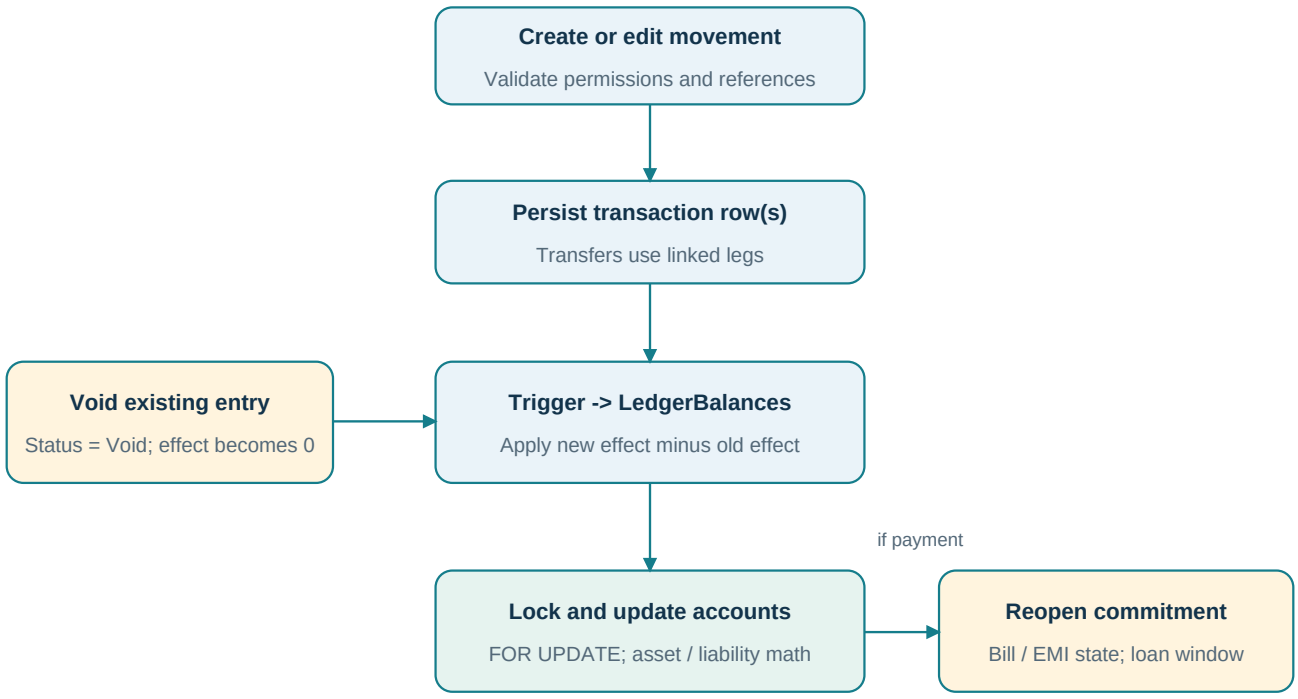


Figure 7. Ledger posting, balance update and void flowchart

Posting and voiding pass through the transaction trigger. Delta math updates locked accounts; payment reversals also restore the corresponding commitment state.

Source: TransactionService.cls; TransactionTrigger.trigger; LedgerBalances.cls; LoanService.afterVoid.

8.1 Balance mechanics

TransactionTrigger invokes LedgerBalances to apply the difference between old and new net effects. Credit contributes +amount, Debit contributes -amount, and Void contributes zero. The handler locks affected accounts FOR UPDATE before updating balances. Incremental deltas keep posting cost independent of total ledger history; recalculate() is the full-ledger recovery path.

Account / operation	Rule	Example
Asset balance	Opening balance + net ledger effect	Bank starts at INR 10,000; an expense debit of INR 1,000 leaves INR 9,000.
Liability balance	Opening amount owed - net ledger effect	Card owes INR 5,000; a purchase debit of INR 1,000 increases owed balance to INR 6,000.
Transfer	Source debit + target credit, linked by group and counter IDs	Bank to wallet changes account balances, without increasing family income or spending.

Account / operation	Rule	Example
Credit-card payment	Linked Card Payment rows	Bank debit and card credit reduce cash and card debt; purchase expense was already recognized.
Reconcile to statement	Post an Adjustment explaining the difference	Use Accounts -> Match to statement instead of manually overwriting the current balance.
Void	Set status to Void; trigger removes the prior net effect	History remains visible while balances and commitment status are corrected.

8.2 Transactions and validation

Manual entry accepts Income, Expense and Refund; transfers use TransferService. Services validate account usability, category type, member attribution, amount and transaction date. Admins can attribute entries to a family member; other roles can post only for themselves. Transfer source must be a usable non-liability account, source and destination must differ, and destination must be active and in the family. General transfers reject Loan targets; loan payments use LoanService.

Transaction edits update both transfer legs together for supported fields. Changing a transfer's accounts requires voiding and recreating it. Bill and EMI payments cannot be edited; void them and pay again. Voiding linked payment groups restores bill or installment state and rebuilds the loan window as needed.

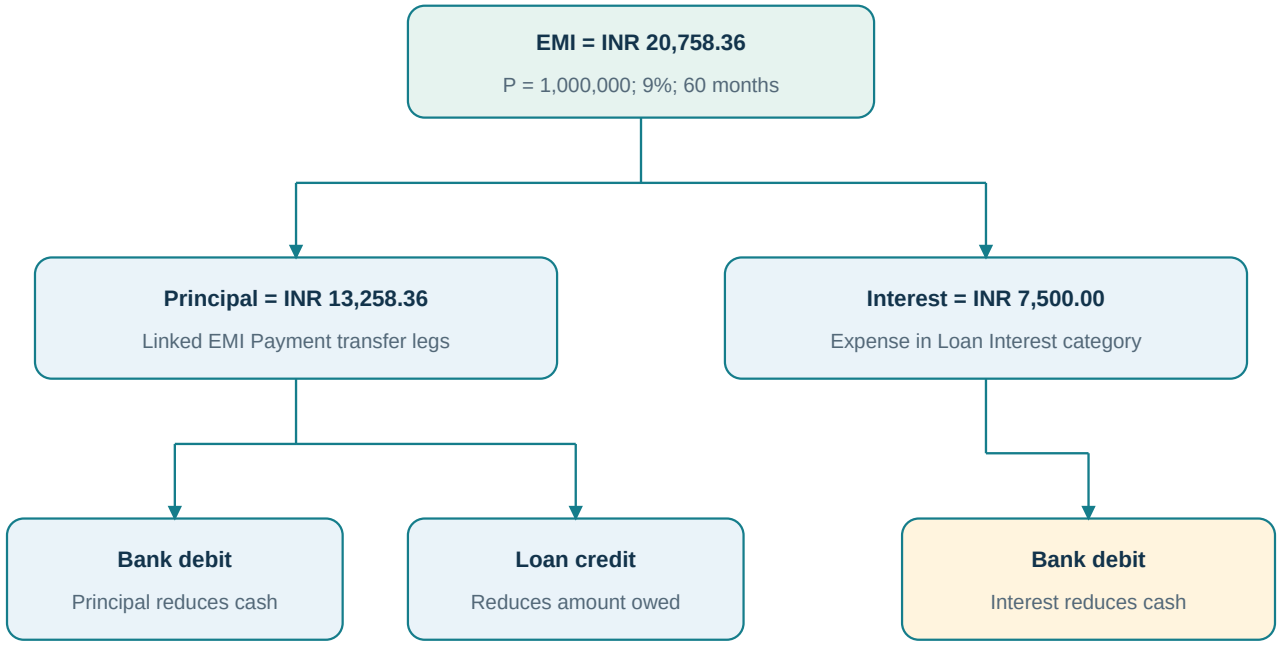
The transaction list supports month or from/to dates, types, account, category including children, member where allowed, search text and optional void records. Default page size is 50 with a maximum of 200. Pagination uses a date/name cursor; aggregate totals are returned on the initial page. This avoids deep OFFSET pagination.

8.3 Categories, budgets and dashboard

Category templates are copied into family-specific records. Parent-child categories support drilldown and aggregation, while system categories identify transfers and card payments. A monthly Budget__c groups Budget_Line__c planned category amounts. BudgetService computes actuals from ledger activity and supports copying the prior month. DashboardService combines scoped balances and monthly figures; InsightService stores each person's overview layout.

Source: LedgerBalances.cls; TransactionTrigger.trigger; TransactionService.cls; TransferService.cls; AccountService.cls; CategoryService.cls; BudgetService.cls; DashboardService.cls.

9. Bills, loans, cards and recurring commitments



First-installment example. Principal is debt reduction; only interest is expense. The bank loses the full EMI.

Figure 8. EMI payment accounting diagram

One installment becomes a principal pair plus an interest expense when interest is positive. This preserves debt reduction separately from family spending.

Source: EmiCalculator.cls; LoanService.payEmi and principalLegs.

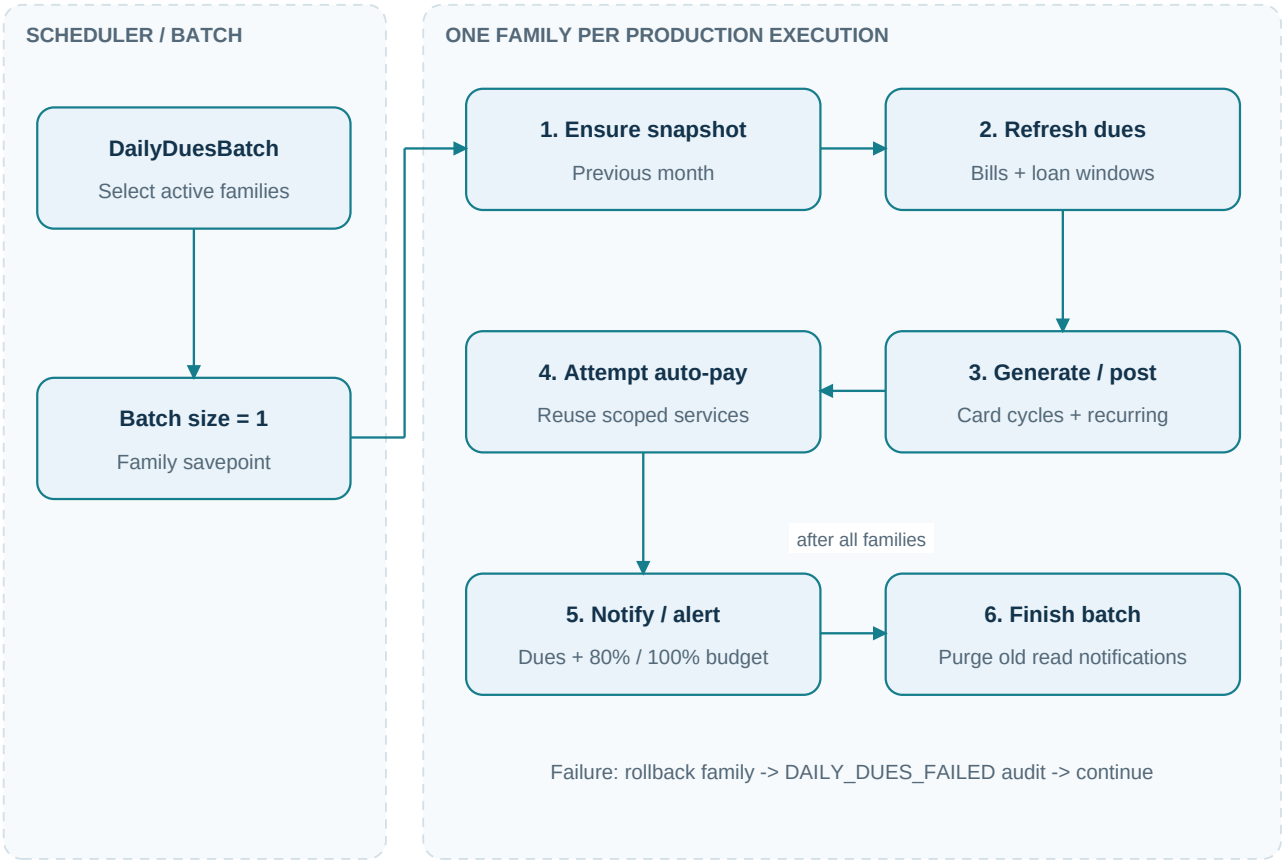
Module	State / calculation	Payment and reversal behavior
Bills	Bill definition produces dated Bill_Instance records; refresh derives due/overdue state and tracks amount due/paid.	Pay creates Bill Payment ledger activity; skip is explicit; voiding payment reopens the corresponding obligation.
Loans / EMIs	EmiCalculator computes amortization; only the next 12 open installments are persisted, plus historical paid records.	EMI splits principal transfer legs from interest expense. Payment updates paid count and rebuilds the next installment window.
Loan prepayment	Principal-only payment, no greater than outstanding.	EMI remains the same; remaining tenure shortens. Ledger reversal rebuilds future installments.
Cards	Statements snapshot cycle-end owed amount, minimum due and due date; unique cycle keys prevent duplicate creation.	Card Payment/Adjustment credits determine payment progress. The card is paid through linked ledger movements.
Recurring entries	Weekly, Monthly, Quarterly or Yearly; next-run date advances after post/skip; MAX_CATCH_UP = 3.	Configured auto-post reuses normal transaction/transfer services; manual confirmation can post or skip.

9.1 EMI calculation and accounting example

For principal P, monthly rate r = annual percentage / 1200 and tenure n, EMI = P*r*(1+r)^n / ((1+r)^n - 1). Zero-rate loans use P/n. Each period rounds interest to two decimal places, principal is EMI minus interest, and the final payment absorbs the remaining principal. An EMI too small to repay the balance is rejected.

For INR 1,000,000 at 9% for 60 months, the calculated EMI is approximately INR 20,758.36. First-month interest is INR 7,500.00 and principal is INR 13,258.36. LoanService records the principal as linked movements from the paying account to the loan account and the interest as an Expense from the paying account. This keeps principal reduction separate from expense reporting. The full computed schedule is capped at 600 rows in current service calls.

9.2 Daily processing and reruns



Unique cycle keys, rule advancement and recipient dedupe keys support reruns; review concurrency separately.

Figure 9. Daily automation activity diagram

Batch execution isolates each family. The finish-stage notification purge runs after the batch; a family failure is rolled back and audited without cancelling other families.

Source: DailyDuesBatch.cls; DuesProcessor.run; NotificationService.cls.

DuesProcessor first ensures the prior monthly snapshot, refreshes bill instances and loan windows, generates card statements and processes recurring rules. It then attempts configured bill/EMI auto-pay, sends bill/EMI/card reminders and evaluates budget thresholds. Automation builds a real member context through SystemContext so the reused services still apply role and scope checks.

DailyDuesBatch uses one family per production batch execution. Each family has a savepoint; an exception rolls back that family's work and emits DAILY_DUES_FAILED. Completion purges read notifications beyond retention. Unique keys, due-state advancement and notification dedupe keys support reruns; they should not be interpreted as a general distributed exactly-once guarantee for every concurrent payment request.

Bill/EMI reminder buckets include 7, 3, 1 and 0 days; card reminders inspect a five-day horizon. Budget notifications go to admins at 80% and 100%. Dedupe keys include family, event identity and recipient. Auto-pay records bookkeeping movements inside Salesforce; it does not initiate an external bank payment.

Source: BillService.cls; LoanService.cls; EmiCalculator.cls; CardStatementService.cls; RecurringService.cls; DuesProcessor.cls; DailyDuesBatch.cls; SystemContext.cls.

10. Reports, snapshots and export

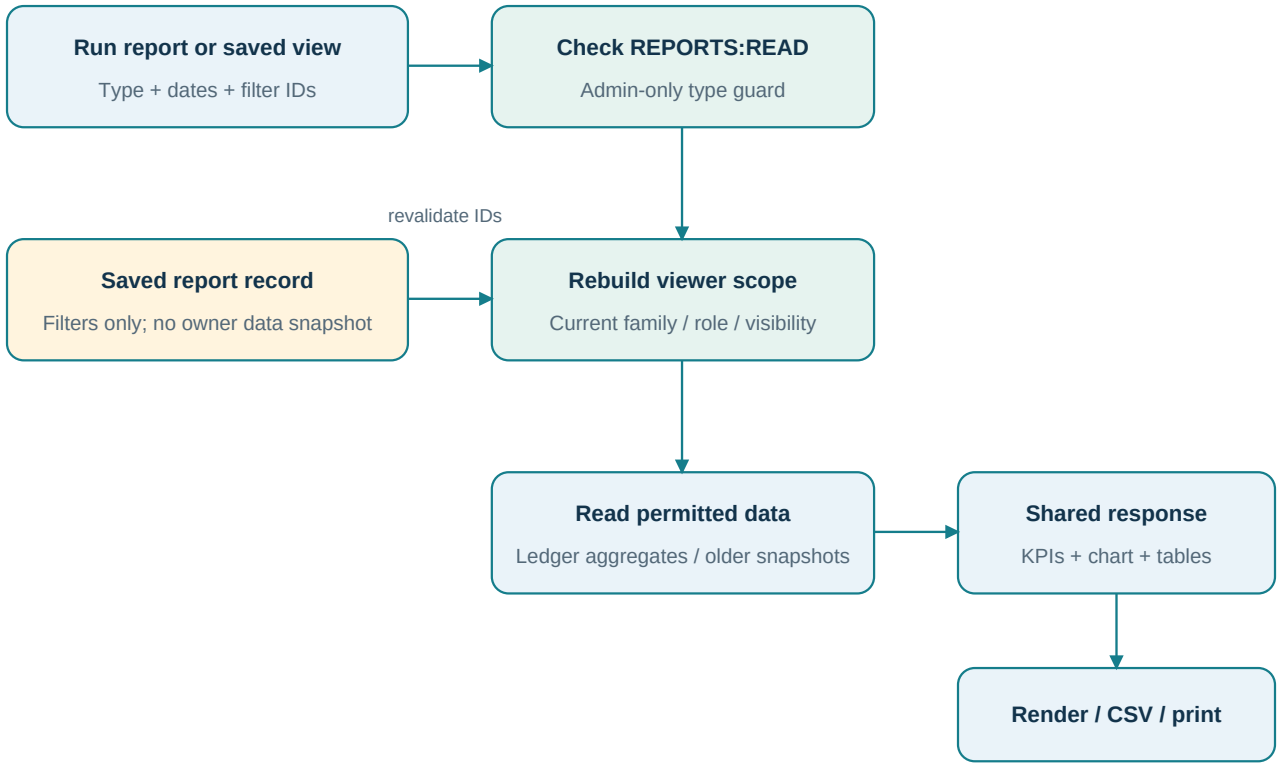


Figure 10. Report execution and saved-filter data flow

A shared report definition never grants the owner's data access. Filters are resolved again and the report is read using the current viewer's family and visibility.

Source: ReportService.cls; InsightService.cls; ReportView.tsx.

ReportService uses a shared response shape: type/title/subtitle, date range, KPI list, one chart, a primary table, optional detail table, note and drilldown information. ReportView renders that contract across report types. Default period is the selected/current month; the service rejects reversed ranges and spans beyond MAX_DAYS = 1,100. Member, category and account filters are revalidated against the caller's scope.

Type	Report	Access	Purpose
summary	Monthly summary	REPORTS:READ + scope	Income, spending and net position.
category	Category spending	REPORTS:READ + scope	Category and subcategory drilldown.
members	Member comparison	Admins only	Family member attribution and comparison.
trend	Spending / income trend	REPORTS:READ + scope	Historical month series, using snapshots for older months where applicable.
budget	Budget versus actual	REPORTS:READ + scope	Planned allocations compared with ledger actuals.
forecast	Cash-flow forecast	Admins only	Upcoming commitments and available liquid funds.
account	Account view	REPORTS:READ + scope	Scoped account activity and balances.

Type	Report	Access	Purpose
cards	Credit cards	REPORTS:READ + scope	Card balances, utilization and payment view.
loans	Loan view	REPORTS:READ + scope	Outstanding amounts and installment context.
bills	Bills view	Admins only	Family obligations and payment state.
modes	Payment modes	REPORTS:READ + scope	Spending grouped by payment mode.
annual	Annual summary	Admins only	Family year-level comparison.
security	Security activity	Admins only	Family security/audit view.

Saved_Report__c stores report type and filter configuration rather than a frozen data extract. Sharing a saved report does not share the owner's access: the report is rerun using the viewer's current scope. Dashboard widget order/visibility is saved in Portal_User__c.Dashboard_Layout__c. Monthly_Summary__c snapshots support trend months older than three; the daily job ensures the previous month exists.

CSV export and browser print are portal capabilities. The Print workflow can be saved as PDF by the browser. These are separate from the DOCX/PDF project documentation generated with this guide.

Source: ReportService.cls; InsightService.cls; src/features/reports/ReportView.tsx; src/api/reportTypes.ts.

11. Portal implementation and API contract

11.1 Frontend organization

Area	Implementation
Routing	routes.tsx: public login/forgot; reset-password and first-login; authenticated AppLayout with feature guards.
Lazy loading	Pages beyond the overview are loaded on demand using React.lazy.
Authentication state	AuthContext, RouteGuards and SessionTimeout coordinate session state and sign-in transitions.
Shared finance state	LedgerContext and typed API DTOs support common account/category/transaction interactions.
API adapter	vkfClient caches Data SDK initialization, attaches session header and converts errors to ApiError.
Session expiration	A 401 SESSION_EXPIRED with a token clears storage and dispatches vkf:session-expired.
Feature pages	Accounts, transactions, categories, budgets, bills, loans, recurring, family, profile/security, reports and help.
UI system	Tailwind styling, shared UI primitives, finance components, charts, QR and OTP controls.

11.2 REST resource map

Base path: /services/apexrest/vkf/v1. The mappings below are read directly from Apex annotations. Wildcard resources dispatch child paths internally. Most protected resources require a full session; login/reset/MFA and ping have flow-specific authentication rules.

Resource class	Mapping (after /services/apexrest)	HTTP handlers
VkfAccountsResource	/vkf/v1/accounts/*	Get, Post, Patch
VkfAuthResource	/vkf/v1/auth/*	Post, Get, Put
VkfBillsResource	/vkf/v1/bills/*	Get, Post, Patch
VkfBudgetsResource	/vkf/v1/budgets/*	Get, Put, Post
VkfCategoriesResource	/vkf/v1/categories/*	Get, Post, Patch
VkfDashboardResource	/vkf/v1/dashboard/*	Get
VkfFamilyResource	/vkf/v1/family/*	Get, Post, Patch, Delete
VkfLoansResource	/vkf/v1/loans/*	Get, Post, Patch
VkfNotificationsResource	/vkf/v1/notifications/*	Get, Post
VkfPingResource	/vkf/v1/ping	Get
VkfProfileResource	/vkf/v1/profile/*	Get, Post, Delete
VkfRecurringResource	/vkf/v1/recurring/*	Get, Post, Patch

Resource class	Mapping (after /services/apexrest)	HTTP handlers
VkfReportsResource	/vkf/v1/reports/*	Get, Post, Put, Delete
VkfTransactionsResource	/vkf/v1/transactions/*	Get, Post, Patch, Delete
VkfTransfersResource	/vkf/v1/transfers/*	Post

11.3 Client operation catalog

The following operations are extracted from endpoints.ts. {id} represents a URL-encoded client identifier or path parameter; query expressions are omitted from the path column. Request fields and return types are defined in src/api/types.ts, financeTypes.ts, commitmentTypes.ts and reportTypes.ts. Server handlers remain authoritative.

Client operation	Verb	Path
authApi.login	POST	/auth/login
authApi.verifyMfa	POST	/auth/mfa
authApi.logout	POST	/auth/logout
authApi.session	GET	/auth/session
authApi.firstLogin	POST	/auth/first-login
authApi.changePassword	PUT	/auth/password
authApi.forgot	POST	/auth/forgot
authApi.reset	POST	/auth/reset
profileApi.me	GET	/profile/me
profileApi.sessions	GET	/profile/sessions
profileApi.signOutOtherDevices	DELETE	/profile/sessions
profileApi.signOutDevice	DELETE	/profile/sessions/{id}
profileApi.security	GET	/profile/security
profileApi.startMfa	POST	/profile/mfa/setup
profileApi.confirmMfa	POST	/profile/mfa/confirm
profileApi.disableMfa	POST	/profile/mfa/disable
profileApi.renewRecoveryCodes	POST	/profile/mfa/recovery-codes
familyApi.members	GET	/family/members
familyApi.addMember	POST	/family/members
familyApi.updateMember	PATCH	/family/members/{id}
familyApi.unlock	POST	/family/members/{id}/unlock
familyApi.resetPassword	POST	/family/members/{id}/reset-password
familyApi.resetMfa	POST	/family/members/{id}/reset-mfa

Client operation	Verb	Path
familyApi.signOutMember	DELETE	/family/members/{id}/sessions
financeApi.accounts	GET	/accounts
financeApi.transferTargets	GET	/accounts/targets
financeApi.createAccount	POST	/accounts
financeApi.updateAccount	PATCH	/accounts/{id}
financeApi.reconcile	POST	/accounts/{id}/reconcile
financeApi.categories	GET	/categories
financeApi.createCategory	POST	/categories
financeApi.updateCategory	PATCH	/categories/{id}
financeApi.transactions	GET	/transactions
financeApi.createEntry	POST	/transactions
financeApi.updateTransaction	PATCH	/transactions/{id}
financeApi.voidTransaction	DELETE	/transactions/{id}
financeApi.transfer	POST	/transfers
financeApi.budget	GET	/budgets/{id}
financeApi.saveBudget	PUT	/budgets/{id}
financeApi.copyBudget	POST	/budgets/{id}/copy-previous
financeApi.dashboard	GET	/dashboard
commitmentApi.bills	GET	/bills
commitmentApi.createBill	POST	/bills
commitmentApi.updateBill	PATCH	/bills/{id}
commitmentApi.payBill	POST	/bills/instances/{id}/pay
commitmentApi.skipBill	POST	/bills/instances/{id}/skip
commitmentApi.loans	GET	/loans
commitmentApi.schedule	GET	/loans/{id}/schedule
commitmentApi.createLoan	POST	/loans
commitmentApi.updateLoan	PATCH	/loans/{id}
commitmentApi.payEmi	POST	/loans/{id}/pay
commitmentApi.prepay	POST	/loans/{id}/prepay
commitmentApi.statements	GET	/accounts/{id}/statements
commitmentApi.rules	GET	/recurring

Client operation	Verb	Path
commitmentApi.createRule	POST	/recurring
commitmentApi.updateRule	PATCH	/recurring/{id}
commitmentApi.postRule	POST	/recurring/{id}/{post|skip}
commitmentApi.notifications	GET	/notifications
commitmentApi.markRead	POST	/notifications/read
reportApi.run	GET	/reports/{id}
reportApi.saved	GET	/reports/saved
reportApi.save	POST	/reports/saved
reportApi.remove	DELETE	/reports/saved/{id}
reportApi.layout	GET	/reports/layout
reportApi.saveLayout	PUT	/reports/layout

11.4 Errors, status codes and payloads

HTTP status	Application codes / meaning
200 / 201 / 204	Successful response, created entity or empty success according to handler.
400	BAD_REQUEST, MFA_EXPIRED; malformed JSON object or invalid request.
401	INVALID_CREDENTIALS or SESSION_EXPIRED.
403	NOT_AUTHORIZED, PASSWORD_CHANGE_REQUIRED or LOCKED.
404	NOT_FOUND, including out-of-family record references.
409	CONFLICT, such as an already closed commitment.
422	VALIDATION or INVALID_CODE.
429	THROTTLED.
500	UNEXPECTED; generic client message, diagnostics restricted to server logging.

```
Request header: X-VKF-Session: <raw session token>
Content-Type: application/json

Example transaction body (use real in-scope record IDs):
{
  "type": "Expense", "accountId": "<account-id>",
  "categoryId": "<expense-category-id>", "amount": 1250,
  "date": "2026-10-28", "notes": "Groceries"
}

Error contract:
{"code":"NOT_AUTHORIZED","errors":["You don't have access to do that."]}
```

Source: src/routes.tsx; src/api/endpoints.ts; src/api/vkfClient.ts; Vkf*Resource.cls; VkfRest.cls; PortalException.cls.

12. Audit, scheduled jobs and retention

AuditService publishes Audit_Event__e immediately. AuditEventTrigger delegates to AuditLogWriter, which persists Audit_Log__b. Immediate publishing means security evidence can survive a rolled-back business transaction; an event is evidence of the logged action/attempt and must be interpreted with its outcome. AuditService is intentionally pinned to API 66.0 in this repository because of guest platform-event access behavior documented by the project.

Job	Schedule expression	Purpose
VKF Session Cleanup	0 0 * * * ?	Hourly: expire stale sessions and remove old closed sessions/spent token state.
VKF Balance Reconcile	0 0 2 * * ?	Daily 02:00: compare balances with ledger, correct drift and audit changes.
VKF Daily Dues	0 0 6 * * ?	Daily 06:00: commitments, recurring entries, reminders and monthly snapshots.
VKF Retention Purge	0 0 4 ? * SUN	Sunday 04:00: remove stale notifications, tokens, password history and rate windows.
VKF Security Digest	0 0 7 ? * MON	Monday 07:00: email a security/operations summary to the scheduling user.

The schedule script retains jobs whose names already exist, so rerunning it does not update an existing cron expression. When changing schedules, inspect and replace the intended existing jobs. Cron expressions do not embed a family time zone; verify the scheduling Salesforce user's time zone and NextFireTime. Family.Time_Zone__c does not by itself prove family-specific execution times.

Record / concern	Retention or operational behavior
Closed sessions	Default seven days; hourly cleanup also marks stale active sessions expired.
Read notifications	Older than 60 days are purged; unread notifications are not covered by this read-only deletion rule.
Auth tokens	Weekly purge removes used or expired tokens older than one day, with a 5,000-row query cap.
Password history	Retain configured history depth; remove surplus and history belonging to removed members.
Rate limits	Weekly purge deletes windows older than one day, capped at 5,000 per query.
Financial ledger / audit	No general ledger or audit deletion policy is established by these housekeeping rules. Define archival separately if required.

The existing security checklist reports a Developer Edition constraint of 5 MB normal data storage and 15 external emails/day; treat these as the recorded environment assumptions, not a newly checked quota. Monitor actual org storage and email limits before adding families. Several purge paths have row caps and the recurring catch-up has a cap, so review backlogs after downtime or growth.

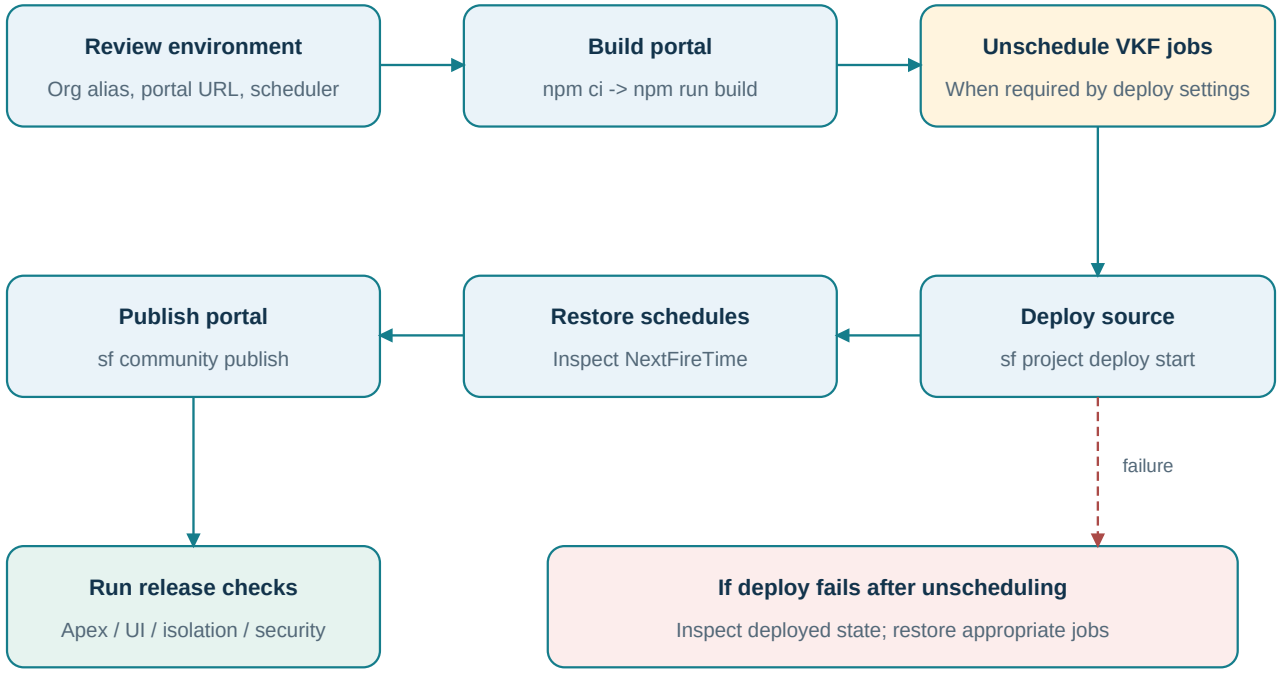
Source: AuditService.cls and metadata; AuditEventTrigger.trigger; AuditLogWriter.cls; schedule-jobs.apex; SessionService.cleanup; RetentionPurgeJob.cls; docs/Security_Checklist.md.

13. Build, configuration and deployment runbook

13.1 Prerequisites and environment settings

Prerequisite	Preparation
Salesforce access	Use the intended org alias, Experience Cloud site and internal deployment user. This guide follows the repository alias VKFinance.
Platform owner	Assign VKF_Platform_Admin permission set for console and administrative scripts.
Local tools	Node >=22, npm and Salesforce CLI; authenticated connection to the intended org.
Portal configuration	Review Security_Policy.Default.Portal_Base_Url__c and Experience Cloud routing for each environment.
Guest API boundary	Preserve VKF_Guest_Api_Access and hardening; do not grant guest family-object CRUD/FLS access.
Scheduler	Confirm scheduler identity, time zone and email destination for the security digest.
Email and recovery	Verify reset email delivery, portal reset links and owner support procedures before onboarding.

13.2 Build and deploy commands



Source rollback does not undo financial data. The document runbook provides exact commands.

Figure 11. Deployment and schedule-recovery flowchart

Build the bundle before source deployment, restore required jobs and publish the portal. If deployment fails after unscheduling, inspect the actual org state before restoring jobs.

Source: README.md; schedule-jobs.apex; unschedule-jobs.apex.

Run commands from the project root in PowerShell. Review the target-org alias before any mutating command. npm ci uses the checked-in portal lockfile. Unscheduling is needed where Salesforce blocks deployments of classes referenced by pending jobs; the repository documents a Deployment Settings alternative. These commands are instructions and were not executed as part of this document task.

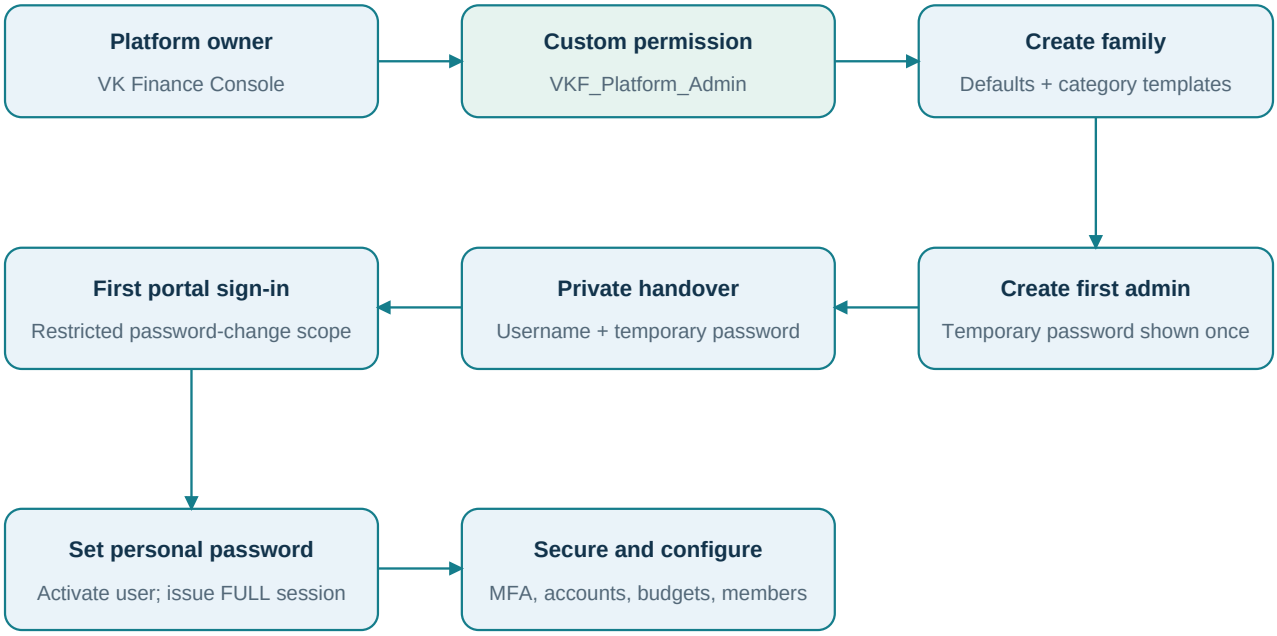
```
# Build portal
Push-Location force-app/main/default/uiBundles/VKFinancePortal
npm ci
npm run build
Pop-Location

# Deploy source, restore schedules and publish the site
sf apex run --target-org VKFinance --file scripts/apex/unschedule-jobs.apex
sf project deploy start --target-org VKFinance --source-dir force-app
sf apex run --target-org VKFinance --file scripts/apex/schedule-jobs.apex
sf community publish --name VKFinancePortal --target-org VKFinance

# Apply the reviewed hardening package
sf project deploy start --target-org VKFinance --metadata-dir config/hardening
```

If deployment fails after jobs are unscheduled, assess the deployed state and restore appropriate schedules using the reviewed script. Publishing and Apex tests are separate steps. Keep a prior known-good source revision and build artifact for recovery; redeploying source does not reverse data mutations or financial entries.

13.3 Onboarding and demo preparation



Member invitations use the same temporary-password principle. Credentials are handed over privately.

Figure 12. Family onboarding and first-access workflow

The platform owner creates the first administrator. Private credential handover leads to a restricted first sign-in, a new password and normal family setup.

Source: VkfConsoleController.onboardFamily; FamilyOnboardingService.cls; AuthService.firstLogin.

Step	Action / expected result
1	Open VK Finance Console -> Onboard Family as an authorized internal user.
2	Enter family and first Family Admin details; inspect currency, time zone and member cap.
3	Share the one-time temporary password privately; first sign-in requires a personal password.
4	Enable MFA for administrators; save recovery codes; add members and assign minimal roles.
5	Add accounts with opening balances, confirm category templates and configure budgets/commitments.
6	For an isolated demo only, use seed-demo-families.apex, seed-demo-finance.apex and seed-demo-commitments.apex (the README says to run the last twice).

Demo credentials are intentionally not reproduced here. Use the existing Demo_Access.md only in the appropriate trusted environment. Demo scripts change records and are not part of a normal production deployment.

Source: README.md; scripts/apex/*.apex; config/hardening; VKF_Platform_Admin.permissionset-meta.xml; VKF_Guest_Api_Access.permissionset-meta.xml.

14. Verification strategy and existing evidence

This documentation task validates generated artifacts and repository-derived tables. It does not certify runtime correctness. The figures below are recorded claims from docs/Security_Checklist.md dated 28 October 2026, not test results produced during this task.

Existing evidence	Recorded result	Interpretation
Live black-box security checks	20 / 20 passed	Historical published-site checks of isolation, auth, guest data access and response hygiene.
Apex tests	All pass; 93% org-wide coverage	Historical org test run; this guide does not include a newly generated test report.
Browser MFA checks	8 / 8 passed	Historical setup, sign-in and admin reset verification.
Code Analyzer security review	58 findings triaged; no open VK issues reported	Includes deliberate system-mode access and template-code findings; review the original rationale.
Salesforce Health Check	85, improved from 64	Remaining Salesforce login/configuration choices are listed in the security checklist.

14.1 Test coverage map

Concern	Representative repository tests	Release expectation
Identity and password rules	AuthServiceTest, PasswordServiceTest, SessionServiceTest	Reject wrong/expired/revoked credentials and enforce password history and restricted sessions.
MFA / security jobs	MfaServiceTest, SecurityJobsTest	Check TOTP vectors, replay, recovery, lockout and housekeeping behavior.
Family and finance isolation	FamilyIsolationTest, FinanceIsolationTest	Use another family's IDs on reads and writes; require 404 and audit behavior.
Ledger / planning	LedgerServiceTest, PlanningServiceTest	Verify debit/credit/liability math, paired transfers, voids, scope and budget totals.
Commitments / automation	CommitmentServiceTest, DuesAutomationTest	Verify payment splits, reopen behavior, reruns, dedupe and due-date changes.
REST / console	VkfResourcesTest, VkfLedgerResourcesTest, VkfCommitmentResourcesTest, VkfConsoleControllerTest	Verify dispatch, status codes, safe payloads and internal authorization.
Insights	ReportServiceTest	Verify report scope, admin-only types, aggregates and saved-filter behavior.
Portal unit / browser tooling	src/api/vkfClient.test.ts; src/lib/finance.test.ts; e2e/	Run available portal checks; inspect browser paths for affected features.

```
# Apex validation
sf apex run test --target-org VKFinance --test-level RunLocalTests --code-coverage --wait 30

# Portal validation
Push-Location force-app/main/default/uiBundles/VKFinancePortal
npx vitest run
npm run build
Pop-Location
```

```
# Security static analysis
sf code-analyzer run --workspace force-app/main/default/classes --workspace force-
app/main/default/triggers --rule-selector Security

# Live checks require environment-specific test accounts and variables.
# Follow docs/Security_Checklist.md, then run:
node scripts/security/pentest.mjs
```

Release acceptance should include: a fresh login and first-login path; MFA and recovery; a cross-family denial; Member/Child/Viewer restrictions; posting and voiding a transfer; a bill and EMI payment/reversal; budget actuals; report scope; and scheduled-job status. Use dedicated test data and check both UI and API outcomes.

15. Troubleshooting, limitations and handover

Symptom / constraint	Likely cause	Action
Guest API cannot read family objects	Expected absence of direct guest CRUD/FLS	Use the custom REST route and VK session; preserve application checks.
Apex deployment blocked by jobs	Scheduled jobs reference edited classes	Follow unschedule/deploy/reschedule or the reviewed Deployment Settings alternative.
Audit read fails after writes	Big-object query ordering restriction documented by project	Perform audit reads before DML; keep session activity deferred.
Locked account returned as 403	PortalException maps LOCKED to supported status	Use application code to distinguish lockout from a general access denial.
MFA codes consistently fail	Clock skew, replay or expired challenge	Check automatic device time; restart sign-in; use a recovery code if necessary.
Duplicate or missing dues after outage	Backlog, due state or capped catch-up	Review rule dates, unique keys, job failures and catch-up bounds before rerunning.
Balance drift	Bypassed service/trigger path or imported opening data	Inspect ledger and reconcile job audit; correct through approved adjustments or recalculation.
401 background UI API calls	README identifies SDK csrf/telemetry guest calls	Distinguish them from failed VK REST requests; inspect the actual request path.
Data or email quota pressure	Developer Edition scale assumptions	Inspect live limits, retention backlog, unread alerts and user growth.

15.1 Implementation cautions

Do not raise AuditService's API version without revalidating guest audit publishing. Preserve explicit WITH SYSTEM_MODE queries and as system DML only behind authorization. Anonymous Apex scripts follow a different access context and the project explicitly warns against copying those system-mode clauses into anonymous scripts. Avoid variable names that shadow Apex identifiers because Apex is case-insensitive.

General TransferService still contains a legacy validation message describing loan repayment as a later-release feature, although LoanService implements EMI and prepayment. Treat the route behavior as authoritative and consider updating that message during future product maintenance. Loan creation initializes its liability account balance; ongoing balance corrections belong to ledger movements and reconciliation.

Configured family time zones and currency fields should be reviewed before broader geographic rollout: daily processing uses Date.today(), cron follows the scheduler context, and reminder money text is INR-specific. The checked-in project does not establish disaster-recovery objectives, automated backup/restore evidence or an external archival policy. Define and verify those operational requirements before relying on them.

15.2 Handover checklist

Owner	Required handover item
Platform owner	Org/site ownership, VKF_Platform_Admin assignment, scheduler identity and verified next-run times.
Security owner	Admin MFA/recovery, guest access review, open Health Check decisions and fresh release security evidence.

Owner	Required handover item
Developer	Build/deploy process, API 66 audit exception, system-mode boundary, tenant checks and typed DTO contracts.
Finance support	Opening balances, adjustment policy, payment void/repost behavior and reconciliation review.
Operations	Storage/email monitoring, failed jobs, purge backlog, snapshot freshness and backup/restore procedure.
Tester	Two-family test dataset, all five roles and fresh unit/integration/browser/security results.

Appendix A. Complete custom-field dictionary

This appendix is generated from every checked-in objects/*/fields/*.field-meta.xml file. It lists custom fields only; standard Salesforce fields such as Id, Name, CreatedDate and system audit fields are omitted. Required means the metadata required flag, not every conditional requirement enforced by Apex. Relationship targets, formulas, enumerated values and uniqueness are included where present. No runtime records or credential values are included.

Audit_Event__e

Immediate-publish event transporting audit facts to the writer.

Field API name	Type	Req.	Definition / constraints
Actor_Id__c	Text	-	Length 18
Details__c	LongTextArea	-	Length 32768
Event_Type__c	Text	-	Length 40
Family_Id__c	Text	-	Length 18
Object_Name__c	Text	-	Length 80
Occurred_At__c	DateTime	-	Occurred At
Record_Id__c	Text	-	Length 18

Audit_Log__b

Indexed audit history held outside normal custom-object storage.

Field API name	Type	Req.	Definition / constraints
Actor_Id__c	Text	-	Length 18
Details__c	LongTextArea	-	Length 32768
Event_Key__c	Text	Yes	Length 36
Event_Time__c	DateTime	Yes	Event Time
Event_Type__c	Text	-	Length 40
Family_Id__c	Text	Yes	Length 18
Object_Name__c	Text	-	Length 80
Record_Id__c	Text	-	Length 18

Auth_Token__c

Hashed one-time reset, MFA challenge and recovery-token state.

Field API name	Type	Req.	Definition / constraints
Attempts__c	Number	-	Precision 3, scale 0; Default: 0

Field API name	Type	Req.	Definition / constraints
Expires_At__c	DateTime	-	Expires At
Family__c	Lookup	-	Target: Family__c
Portal_User__c	MasterDetail	-	Target: Portal_User__c
Token_Hash__c	Text	Yes	Length 64; Unique; External ID
Type__c	Picklist	-	Values: Reset, Invite, Email OTP, MFA Challenge, Recovery Code
Used_At__c	DateTime	-	Used At

Bill__c

Reusable bill definition, responsible person and payment/reminder configuration.

Field API name	Type	Req.	Definition / constraints
Active__c	Checkbox	-	Default: true
Auto_Pay__c	Checkbox	-	Default: false
Category__c	Lookup	-	Target: Category__c
Consumer_No__c	Text	-	Length 40
Expected_Amount__c	Currency	-	Precision 16, scale 2
Family__c	Lookup	Yes	Target: Family__c
Frequency__c	Picklist	-	Values: Monthly, Bi-monthly, Quarterly, Half-yearly, Yearly, One-time
Next_Due_Date__c	Date	-	Next Due Date
Pay_From_Account__c	Lookup	-	Target: Financial_Account__c
Reminder_Days__c	Number	-	Precision 2, scale 0
Responsible_Member__c	Lookup	-	Target: Portal_User__c

Bill_Instance__c

Dated obligation with amounts due, paid and current status.

Field API name	Type	Req.	Definition / constraints
Amount_Due__c	Currency	-	Precision 16, scale 2
Amount_Paid__c	Currency	-	Precision 16, scale 2
Bill__c	MasterDetail	-	Target: Bill__c
Due_Date__c	Date	Yes	Due Date
Family__c	Lookup	Yes	Target: Family__c

Field API name	Type	Req.	Definition / constraints
Paid_On__c	Date	-	Paid On
Status__c	Picklist	-	Values: Upcoming, Due, Overdue, Partially Paid, Paid, Skipped
Unique_Key__c	Text	-	Length 60; Unique; External ID

Budget__c

Monthly family budget header.

Field API name	Type	Req.	Definition / constraints
Family__c	Lookup	Yes	Target: Family__c
Month__c	Date	Yes	First day of the month.
Notes__c	Text	-	Length 255
Unique_Key__c	Text	-	Length 60; Unique; External ID; Family Id + month; one budget per family per month.

Budget_Line__c

Category allocation belonging to a budget.

Field API name	Type	Req.	Definition / constraints
Budget__c	MasterDetail	-	Target: Budget__c
Category__c	Lookup	Yes	Target: Category__c
Family__c	Lookup	Yes	Target: Family__c
Planned__c	Currency	-	Precision 16, scale 2

Card_Statement__c

Card cycle snapshot, statement amount, minimum payment and due date.

Field API name	Type	Req.	Definition / constraints
Account__c	Lookup	Yes	Target: Financial_Account__c
Cycle_End__c	Date	Yes	Cycle End
Cycle_Start__c	Date	-	Cycle Start
Due_Date__c	Date	-	Due Date
Family__c	Lookup	Yes	Target: Family__c
Minimum_Due__c	Currency	-	Precision 16, scale 2
Statement_Amount__c	Currency	-	Precision 16, scale 2

Field API name	Type	Req.	Definition / constraints
Unique_Key__c	Text	-	Length 60; Unique; External ID

Category__c

Family-specific category hierarchy, kinds and system categories.

Field API name	Type	Req.	Definition / constraints
Active__c	Checkbox	-	Default: true
Color__c	Text	-	Length 20
Family__c	Lookup	Yes	Target: Family__c
Icon__c	Text	-	Length 40
Is_Essential__c	Checkbox	-	Default: false
Kind__c	Picklist	-	Values: Expense, Income, System
Parent_Category__c	Lookup	-	Target: Category__c
Sort_Order__c	Number	-	Precision 4, scale 0
System_Key__c	Text	-	Length 40
Tax_Relevant__c	Checkbox	-	Default: false

Default_Category__mdt

Category template copied into each family.

Field API name	Type	Req.	Definition / constraints
Color__c	Text	-	Length 20
Icon__c	Text	-	Length 40
Is_Essential__c	Checkbox	-	Default: false
Kind__c	Text	-	Length 20
Parent_Key__c	Text	-	Length 40
Sort_Order__c	Number	-	Precision 4, scale 0
System_Key__c	Text	-	Length 40
Tax_Relevant__c	Checkbox	-	Default: false

EMI_Installment__c

Persisted loan installment with principal/interest split and due status.

Field API name	Type	Req.	Definition / constraints
Closing_Balance__c	Currency	-	Precision 16, scale 2
Due_Date__c	Date	Yes	Due Date
EMI_Amount__c	Currency	-	Precision 16, scale 2
Family__c	Lookup	Yes	Target: Family__c
Installment_No__c	Number	-	Precision 4, scale 0
Interest_Part__c	Currency	-	Precision 16, scale 2
Loan__c	MasterDetail	-	Target: Loan__c
Opening_Balance__c	Currency	-	Precision 16, scale 2
Paid_On__c	Date	-	Paid On
Principal_Part__c	Currency	-	Precision 16, scale 2
Status__c	Picklist	-	Values: Upcoming, Due, Overdue, Paid, Skipped
Unique_Key__c	Text	-	Length 60; Unique; External ID

Family__c

Tenant root; family configuration, status, currency, time zone and member cap.

Field API name	Type	Req.	Definition / constraints
Base_Currency__c	Text	-	Length 3; Default: "INR"
Family_Code__c	AutoNumber	-	External ID
Large_Txn_Threshold__c	Currency	-	Precision 16, scale 2
Max_Members__c	Number	-	Precision 3, scale 0; Default: 10
Status__c	Picklist	-	Values: Active, Suspended
Time_Zone__c	Text	-	Length 60; Default: "Asia/Kolkata"

Financial_Account__c

Bank, cash, wallet, card, loan and investment balances and ownership.

Field API name	Type	Req.	Definition / constraints
Color__c	Text	-	Length 20
Credit_Limit__c	Currency	-	Precision 16, scale 2
Current_Balance__c	Currency	-	Precision 16, scale 2; Maintained by TransactionTrigger. Assets: money held. Liabilities: money owed.
Family__c	Lookup	Yes	Target: Family__c

Field API name	Type	Req.	Definition / constraints
Institution__c	Text	-	Length 80
Is_Liability__c	Checkbox	-	Formula: OR(ISPICKVAL(Type__c, 'Credit Card'), ISPICKVAL(Type__c, 'Loan')); Credit cards and loans: the balance is money owed.
Is_Shared__c	Checkbox	-	Default: false
Last4__c	Text	-	Length 4; Never store full account or card numbers.
Min_Due_Pct__c	Number	-	Precision 5, scale 2
Opening_Balance__c	Currency	-	Precision 16, scale 2; For cards and loans: outstanding owed.
Opening_Date__c	Date	-	Opening Date
Owner_Member__c	Lookup	-	Target: Portal_User__c
Payment_Due_Days__c	Number	-	Precision 2, scale 0
Sort_Order__c	Number	-	Precision 4, scale 0
Statement_Day__c	Number	-	Precision 2, scale 0
Status__c	Picklist	-	Values: Active, Closed
Type__c	Picklist	-	Values: Savings, Current, Cash, UPI Wallet, Credit Card, Loan, Fixed Deposit, Investment

Loan__c

Loan terms, borrower, loan account, EMI and outstanding principal.

Field API name	Type	Req.	Definition / constraints
Annual_Rate__c	Number	-	Precision 5, scale 2
Auto_Debit__c	Checkbox	-	Default: false
Borrower_Member__c	Lookup	-	Target: Portal_User__c
Debit_Account__c	Lookup	-	Target: Financial_Account__c
EMI_Amount__c	Currency	-	Precision 16, scale 2
Family__c	Lookup	Yes	Target: Family__c
Installments_Paid__c	Number	-	Precision 4, scale 0
Lender__c	Text	-	Length 80
Loan_Account__c	Lookup	-	Target: Financial_Account__c
Loan_Type__c	Picklist	-	Values: Home, Car, Personal, Education, Gold, Consumer Durable, Card EMI, Other
Outstanding_Principal__c	Currency	-	Precision 16, scale 2
Principal__c	Currency	-	Precision 16, scale 2

Field API name	Type	Req.	Definition / constraints
Start_Date__c	Date	-	First EMI Date
Status__c	Picklist	-	Values: Active, Closed
Tenure_Months__c	Number	-	Precision 4, scale 0

Monthly_Summary__c

Monthly snapshots supporting older historical trends.

Field API name	Type	Req.	Definition / constraints
By_Category__c	LongTextArea	-	Length 32768
Expense__c	Currency	-	Precision 16, scale 2
Family__c	Lookup	Yes	Target: Family__c
Income__c	Currency	-	Precision 16, scale 2
Member__c	Lookup	-	Target: Portal_User__c
Month__c	Date	Yes	Month
Savings__c	Currency	-	Precision 16, scale 2
Unique_Key__c	Text	-	Length 60; Unique; External ID

Notification__c

Recipient-specific alert, severity, read state and deduplication key.

Field API name	Type	Req.	Definition / constraints
Dedupe_Key__c	Text	-	Length 120; Unique; External ID; Stops the daily job from repeating the same alert.
Family__c	Lookup	Yes	Target: Family__c
Is_Read__c	Checkbox	-	Default: false
Link__c	Text	-	Length 255
Message__c	Text	-	Length 255
Recipient__c	Lookup	Yes	Target: Portal_User__c
Severity__c	Picklist	-	Values: info, success, warning, critical
Title__c	Text	-	Length 120
Type__c	Text	-	Length 40

Password_History__c

Previous salted hashes used to enforce password reuse policy.

Field API name	Type	Req.	Definition / constraints
Hash_Iterations__c	Number	-	Precision 7, scale 0
Password_Hash__c	Text	-	Length 100
Password_Salt__c	Text	-	Length 64
Portal_User__c	MasterDetail	-	Target: Portal_User__c

Portal_Session__c

Hashed session token, user/family linkage, scope, activity and revocation.

Field API name	Type	Req.	Definition / constraints
Ended_At__c	DateTime	-	Ended At
Expires_At__c	DateTime	-	Expires At
Family__c	Lookup	-	Target: Family__c
Last_Activity__c	DateTime	-	Last Activity
Portal_User__c	MasterDetail	-	Target: Portal_User__c
Revoke_Reason__c	Text	-	Length 80
Scope__c	Picklist	-	Values: FULL, CHANGE_PASSWORD
Status__c	Picklist	-	Values: Active, Logged Out, Expired, Revoked
Token_Hash__c	Text	Yes	Length 64; Unique; External ID; SHA-256 of the session token. Secret.
User_Agent__c	Text	-	Length 255
User_Agent_Hash__c	Text	-	Length 64

Portal_User__c

Custom portal identity, family membership, role, password state, MFA and dashboard layout.

Field API name	Type	Req.	Definition / constraints
Dashboard_Layout__c	LongTextArea	-	Length 1000
Email__c	Email	-	Email
Failed_Login_Count__c	Number	-	Precision 3, scale 0; Default: 0
Family__c	Lookup	Yes	Target: Family__c
First_Name__c	Text	Yes	Length 40
Hash_Iterations__c	Number	-	Precision 7, scale 0
Last_Login__c	DateTime	-	Last Login

Field API name	Type	Req.	Definition / constraints
Last_Name__c	Text	Yes	Length 80
Locked_Until__c	DateTime	-	Locked Until
Lockout_Count__c	Number	-	Precision 3, scale 0; Default: 0
Login_Window_Attempts__c	Number	-	Precision 4, scale 0; Default: 0
Login_Window_Start__c	DateTime	-	Login Window Start
MFA_Enrolled_At__c	DateTime	-	Two-Step Enrolled At
MFA_Last_Step__c	Number	-	Precision 18, scale 0; Last accepted 30-second code window, so a code can't be replayed.
MFA_Method__c	Picklist	-	Values: None, Authenticator; Two-step verification method. None until an authenticator app is confirmed.
MFA_Secret__c	EncryptedText	-	Length 64; Secret. Authenticator-app key (hex), encrypted at rest. No FLS for any profile; read only by Apex.
Mobile__c	Phone	-	Mobile
Monthly_Spend_Limit__c	Currency	-	Precision 16, scale 2
Must_Change_Password__c	Checkbox	-	Default: false
Password_Changed_At__c	DateTime	-	Password Changed At
Password_Hash__c	Text	-	Length 100; Secret. No FLS for any profile.
Password_Salt__c	Text	-	Length 64; Secret. No FLS for any profile.
Relationship__c	Picklist	-	Values: Self, Spouse, Son, Daughter, Father, Mother, Brother, Sister, Grandparent, Other
Role__c	Picklist	-	Values: Family Admin, Co-Admin, Member, Child, Viewer
Status__c	Picklist	-	Values: Invited, Active, Disabled, Removed
Temp_Password_Expires_At__c	DateTime	-	Temporary Password Expires At
Username__c	Text	Yes	Length 40; Unique; External ID; Globally unique across all families, stored lower case.
Viewer_Scope__c	Picklist	-	Values: Own, Family

Rate_Limit__c

Hashed client-address or throttle bucket and window counters.

Field API name	Type	Req.	Definition / constraints
Hits__c	Number	-	Precision 9, scale 0; Default: 0
Key__c	Text	-	Length 80; Unique; External ID; Bucket name plus a SHA-256 hash of the client address - never the raw address.
Window_Start__c	DateTime	-	Window Start

Recurring_Rule__c

Recurring transaction configuration, cadence and next-run state.

Field API name	Type	Req.	Definition / constraints
Account__c	Lookup	-	Target: Financial_Account__c
Active__c	Checkbox	-	Default: true
Amount__c	Currency	-	Precision 16, scale 2
Category__c	Lookup	-	Target: Category__c
End_Date__c	Date	-	End Date
Family__c	Lookup	Yes	Target: Family__c
Frequency__c	Picklist	-	Values: Weekly, Monthly, Quarterly, Yearly
Last_Run_Date__c	Date	-	Last Run Date
Member__c	Lookup	-	Target: Portal_User__c
Mode__c	Picklist	-	Values: Auto-post, Suggest
Next_Run_Date__c	Date	-	Next Run Date
Payee__c	Text	-	Length 120
Payment_Mode__c	Picklist	-	Values: UPI, Cash, Debit Card, Credit Card, NEFT/IMPS, Net Banking, Cheque, Auto-debit, Other
To_Account__c	Lookup	-	Target: Financial_Account__c
Type__c	Picklist	-	Values: Expense, Income, Transfer

Role_Permission__mdt

Role x feature x action grants plus descriptive scope.

Field API name	Type	Req.	Definition / constraints
Actions__c	Text	-	Length 100
Feature__c	Text	-	Length 40
Role__c	Text	-	Length 40
Scope__c	Text	-	Length 20

Saved_Report__c

Report type and saved filters; private or family shared.

Field API name	Type	Req.	Definition / constraints
Family__c	Lookup	Yes	Target: Family__c

Field API name	Type	Req.	Definition / constraints
Filters__c	LongTextArea	-	Length 4000
Owner_Member__c	Lookup	Yes	Target: Portal_User__c
Report_Type__c	Text	-	Length 40
Shared_With_Family__c	Checkbox	-	Default: false

Security_Policy__mdt

Default timeouts, password, hashing, throttling and portal URL configuration.

Field API name	Type	Req.	Definition / constraints
Absolute_Timeout_Hours__c	Number	-	Precision 18, scale 0
Hash_Iterations__c	Number	-	Precision 18, scale 0
Idle_Timeout_Minutes__c	Number	-	Precision 18, scale 0
Ip_Max_Failures__c	Number	-	Precision 18, scale 0
Lockout_Minutes__c	Number	-	Precision 18, scale 0
Max_Failed_Logins__c	Number	-	Precision 18, scale 0
Max_Sessions_Per_User__c	Number	-	Precision 18, scale 0
MFA_Challenge_Minutes__c	Number	-	Precision 18, scale 0
Min_Admin_Password_Length__c	Number	-	Precision 18, scale 0
Min_Password_Length__c	Number	-	Precision 18, scale 0
Password_History_Count__c	Number	-	Precision 18, scale 0
Portal_Base_Url__c	Text	-	Length 255
Reset_Token_Minutes__c	Number	-	Precision 18, scale 0
Session_Retention_Days__c	Number	-	Precision 18, scale 0
Temp_Password_Hours__c	Number	-	Precision 18, scale 0
Throttle_Max_Attempts__c	Number	-	Precision 18, scale 0
Throttle_Window_Minutes__c	Number	-	Precision 18, scale 0

Transaction__c

One ledger movement; linked transfer/payment rows and commitment references.

Field API name	Type	Req.	Definition / constraints
Account__c	Lookup	Yes	Target: Financial_Account__c
Amount__c	Currency	Yes	Precision 16, scale 2

Field API name	Type	Req.	Definition / constraints
Bill_Instance__c	Lookup	-	Target: Bill_Instance__c
Category__c	Lookup	-	Target: Category__c
Counter_Transaction__c	Lookup	-	Target: Transaction__c
Direction__c	Picklist	-	Values: Debit, Credit
EMI_Installment__c	Lookup	-	Target: EMI_Installment__c
Family__c	Lookup	Yes	Target: Family__c
Member__c	Lookup	-	Target: Portal_User__c
Notes__c	Text	-	Length 255
Payee__c	Text	-	Length 120
Payment_Mode__c	Picklist	-	Values: UPI, Cash, Debit Card, Credit Card, NEFT/IMPS, Net Banking, Cheque, Auto-debit, Other
Recurring_Rule__c	Lookup	-	Target: Recurring_Rule__c
Reference_No__c	Text	-	Length 60
Source__c	Picklist	-	Values: Manual, Recurring, Auto-pay, Import
Status__c	Picklist	-	Values: Cleared, Pending, Void
Transfer_Group_Id__c	Text	-	Length 36; External ID; Shared by both legs of a transfer.
Txn_Date__c	Date	Yes	Date
Type__c	Picklist	-	Values: Income, Expense, Refund, Transfer Out, Transfer In, Card Payment, Bill Payment, EMI Payment, Adjustment

Source: force-app/main/default/objects/**/fields/*.field-meta.xml; standard fields and Apex-only validations are outside this generated inventory.

Appendix B. Role metadata scope reference

Exact Actions__c and Scope__c values are preserved here for traceability. As explained in section 7, Scope__c is descriptive metadata; AccessGuard consumes role, feature and actions only.

Role	Feature	Actions	Scope
Co-Admin	ACCOUNTS	READ,CREATE,EDIT	FAMILY
Co-Admin	BILLS	READ,CREATE,EDIT,PAY	FAMILY
Co-Admin	BUDGETS	READ,CREATE,EDIT	FAMILY
Co-Admin	CATEGORIES	READ	FAMILY
Co-Admin	DASHBOARD	READ	FAMILY
Co-Admin	LOANS	READ,CREATE,EDIT,PAY	FAMILY
Co-Admin	MEMBER_SECURITY	EDIT	FAMILY
Co-Admin	MEMBERS	READ,CREATE,EDIT	FAMILY
Co-Admin	NOTIFICATIONS	READ	OWN
Co-Admin	RECURRING	READ,CREATE,EDIT	FAMILY
Co-Admin	REPORTS	READ,CREATE	FAMILY
Co-Admin	TRANSACTIONS	READ,CREATE,EDIT,DELETE	FAMILY
Child	ACCOUNTS	READ	OWN
Child	CATEGORIES	READ	FAMILY
Child	DASHBOARD	READ	OWN
Child	MEMBERS	READ	OWN
Child	NOTIFICATIONS	READ	OWN
Child	REPORTS	READ,CREATE	OWN
Child	TRANSACTIONS	READ,CREATE,EDIT	OWN
Family Admin	ACCOUNTS	READ,CREATE,EDIT	FAMILY
Family Admin	BILLS	READ,CREATE,EDIT,PAY	FAMILY
Family Admin	BUDGETS	READ,CREATE,EDIT	FAMILY
Family Admin	CATEGORIES	READ,CREATE,EDIT	FAMILY
Family Admin	DASHBOARD	READ	FAMILY
Family Admin	LOANS	READ,CREATE,EDIT,PAY	FAMILY
Family Admin	MEMBER_SECURITY	EDIT	FAMILY
Family Admin	MEMBERS	READ,CREATE,EDIT	FAMILY
Family Admin	NOTIFICATIONS	READ	OWN

Role	Feature	Actions	Scope
Family Admin	RECURRING	READ,CREATE,EDIT	FAMILY
Family Admin	REPORTS	READ,CREATE	FAMILY
Family Admin	TRANSACTIONS	READ,CREATE,EDIT,DELETE	FAMILY
Member	ACCOUNTS	READ,CREATE,EDIT	OWN_SHARED
Member	BILLS	READ,PAY	ASSIGNED
Member	BUDGETS	READ	FAMILY
Member	CATEGORIES	READ	FAMILY
Member	DASHBOARD	READ	OWN_SHARED
Member	LOANS	READ	OWN
Member	MEMBERS	READ	OWN
Member	NOTIFICATIONS	READ	OWN
Member	RECURRING	READ,CREATE,EDIT	OWN
Member	REPORTS	READ,CREATE	OWN_SHARED
Member	TRANSACTIONS	READ,CREATE,EDIT,DELETE	OWN_SHARED
Viewer	BILLS	READ	VIEWER
Viewer	BUDGETS	READ	FAMILY
Viewer	CATEGORIES	READ	FAMILY
Viewer	DASHBOARD	READ	VIEWER
Viewer	LOANS	READ	VIEWER
Viewer	MEMBERS	READ	OWN
Viewer	NOTIFICATIONS	READ	OWN
Viewer	REPORTS	READ	VIEWER
Viewer	TRANSACTIONS	READ	VIEWER

Appendix C. Source references and glossary

Reference	Use in this document
README.md	Project status, architecture summary, commands and operational cautions.
docs/User_Guide.md	Family workflows and user-facing feature descriptions.
docs/Security_Checklist.md	Historical security results, Health Check decisions and rerun guidance.
docs/VKFinance_Solution_Blueprint.pdf	Existing design artifact; consult separately for original solution intent.
classes/AccessGuard.cls; LedgerScope.cls; FamilyScopedSelector.cls	Role actions, in-family visibility and family tenant boundary.
customMetadata/Role_Permission.*; Security_Policy.Default	Generated permission matrix and default policy values.
objects/**; classes/Vkf*Resource.cls; src/api/endpoints.ts	Generated data dictionary, REST mappings and client operation catalog.
scripts/apex/schedule-jobs.apex; unschedule-jobs.apex	Job names, cron expressions and deployment preparation.
scripts/build_implementation_document.py	Narrative source and repeatable DOCX/PDF/Markdown generation.

Term	Meaning
Tenant	One family whose records are isolated from other families.
DTO	Data transfer object exposing approved response fields.
EMI	Equated monthly installment, split into principal and interest.
TOTP	Time-based one-time password generated by an authenticator app.
Step-up	Re-verification of credentials before a sensitive action.
System mode	Elevated Salesforce data access; application guards must supply member authorization.
Dedupe key	Persisted event/recipient identity used to suppress duplicate alerts.
Snapshot	Persisted monthly aggregate supporting historical insight.